Check for updates

Journal of Product Innovation Management

WILEY | pdma Connecting Innovators Worldwide

ORIGINAL ARTICLE OPEN ACCESS

Reframing Through New Minds: How External Experts Unlock Problem Reframing Through Reasoning Logics

Paola Bellis[1] | Claudio Dell'Era[1] | Roberto Verganti[2]

[1]School of Management, Politecnico di Milano, Milano, Italy | [2]House of Innovation, Stockholm School of Economics, Stockholm, Sweden

Correspondence: Paola Bellis (paola.bellis@polimi.it)

Received: 11 April 2023 | **Revised:** 21 July 2025 | **Accepted:** 17 November 2025

Associate Editor: Aric Rindfleisch

Keywords: abduction | deduction | external experts | induction | problem framing | reasoning logic | reframing

ABSTRACT

This study examines how external experts assist organizations in reframing problems by applying different reasoning logics in the early stages of the innovation process. Recent research emphasizes the need to interpret the ambiguity of the innovation environment to reframe problems. External experts can contribute by offering unconventional perspectives that help shift how problems are understood. While studies have explored the role of external experts in problem solving (e.g., ideation and evaluation), few have investigated their involvement in problem reframing. To this end, we conducted an ethnographic study of four strategic innovation projects in which firms collaborated with external experts of varying expertise. We analyze how experts assist innovation teams in moving from hypothetical to renewed frames using five distinct reasoning logics. Early on, experts use (1) reverse deduction and (2) reverse induction to identify common patterns. Upon encountering new perspectives, they rely on (3) explanatory abduction to explore alternative interpretations. This opens a dialogic space for imagination and insights generation through (4) speculative innovative abduction, and (5) moral innovative abduction. We contribute to research on problem reframing in innovation by showing how external experts support reframing not only by introducing new knowledge, but also by reshaping existing knowledge to foster renewed understanding.

1 | Introduction

"Talk is cheap, so we should do a lot of it," said Gustav Söderström, Chief Product, and Technology Officer at Spotify,[1] in a recent talk. At the beginning of every innovation effort, Spotify spends "even six or seven months debating" to "create some model of the world, to have some hypotheses for where the world is going." These words underscore not only the importance that organizations place on properly framing their innovation projects in novel and meaningful ways, but also the critical role of dialogue and reasoning in the reframing process.

In today's complex and rapidly changing environment, innovators often encounter ambiguous or ill-defined problem spaces with high stakes for the organization's long-term success (Garbuio and Lin 2021). As Ananth et al. (2024) note, tackling complex and ambiguous problems requires individuals and organizations to focus on cultivating a thorough and nuanced understanding of these problems before pursuing solutions. Therefore, understanding the context, identifying changes, and recognizing what is meaningful in the new scenario are essential to define the boundaries of interest and guide the organization's subsequent development (Bouschery et al. 2022; Verganti 2017). In concrete terms, these efforts take shape through framing activities that help clarify and articulate what constitutes the problem at hand. In management literature, frames are defined as "knowledge structures that help individuals to organize and interpret incoming perceptual information by fitting it into already available cognitive representations from memory" (Cornelissen and Werner 2014,

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2026 The Author(s). *Journal of Product Innovation Management* published by Wiley Periodicals LLC on behalf of Product Development & Management Association.

Managerial Summary

This study offers valuable insights for managers. First, it emphasizes that the value of external experts lies not only in their knowledge, but also in how they reason and interact throughout problem reframing. Managers should go beyond focusing on breakthrough ideas and engage in shared reasoning, which fosters more robust and innovative problem reframing. External experts bring complementary perspectives, but it is through collaboration that renewed understanding truly emerges. Second, the study highlights the need for managerial openness to diverse—and sometimes uncomfortable—perspectives. Problem reframing is not just about hearing new views but about rethinking and, at times, letting go of existing assumptions. Because of their distance from the internal firm's logics, external experts might introduce reasoning that challenges dominant mental models. Managers should avoid seeking only validation and instead engage with tensions and disagreements to uncover what was previously overlooked. Finally, the study highlights the transformative power of dialogue. Genuine dialogue, grounded in openness and curiosity, supports deep reflection and enables the co-construction of new frames. Engaging with tensions and disagreements can uncover what was previously overlooked. In this way, feedback becomes a generative force, and problem reframing evolves through continuous learning.

187), while framing refers to the act of "deploying frames to select aspects of perceived reality and make them more salient" (Giorgi 2017, 11). Frames are not static entities and may need to be redefined as the context evolves. This can occur either because the situation that originally shaped a frame is challenged, or because an innovation project is about to start and a first interpretation of the emerging context must be developed. In both cases, ambiguity is likely to re-emerge, requiring the exploration of alternative interpretations, thereby enabling reframing (Bianchi and Verganti 2021). Reynolds et al. (2023) define reframing as "the process of framing innovation in an alternative way to existing framings by introducing new frames or redirecting audience attention to alternative frames" (199). Therefore, problem reframing is defined as the process by which individuals and organizations make sense of the ambiguity of the innovation environment to find new meaningful ways of formulating the problem (Beckman 2020).

Recent studies highlight that one of the key challenges in the pursuit of breakthrough innovation is not only discovering and framing new problems, but reframing them (Magistretti et al. 2025), emphasizing that problem reframing is a fundamental step in laying the groundwork for innovation (Beckman and Barry 2009; Vaccaro et al. 2011). While problem reframing is recognized as critical to fostering innovation (Abdulla et al. 2020; Pham et al. 2023; Unsworth 2001), much of the research has focused on generating solutions to a given problem (Amabile 1988; Kaplan and Vakili 2015) or evaluating those solutions (Berg 2019; Mueller et al. 2014). Recent studies calling to investigate how innovation problems are framed and reframed, and not just how they are solved, have led to a resurgence of interest in problem reframing in the innovation community (Ananth et al. 2024; Posen et al. 2018).

Since problem reframing involves changing an existing way of seeing the problem, an effective approach to reframe an innovation problem is to look at it from different perspectives (Bianchi and Verganti 2021; Rosenbloom et al. 2016; Stigliani and Ravasi 2012). Several innovation studies suggest that individuals from different functional areas, such as marketing, R&D, manufacturing, engineering, purchasing, can provide these different perspectives through cross-functional teams (Slater et al. 2014; Tsai 2023) or multi-party collaborations that extend beyond the individual firm (Barczak et al. 2009; Fjeldstad et al. 2012). Other studies have investigated the relevance of involving external experts, focusing on ideation (e.g., Chesbrough 2003; Chesbrough and Bogers 2014), idea evaluation (e.g., Hammedi et al. 2011; Sukhov et al. 2021) or complex knowledge-intensive problems (Fisher et al. 2018). However, few studies have adequately investigated the contribution of external experts to problem reframing in early stages of the innovation process (e.g., Jepsen et al. 2014; Verganti 2009). Experts have the ability to view a problem from more diverse perspectives, unencumbered by organizational frames (Ainamo et al. 2021; Altuna et al. 2017; Verganti and Shani 2016). When an organization's willingness to change goes beyond finding new solutions to existing problems and instead aims to reframe the challenge and imagine new possibilities, external experts can be particularly valuable. They help to broaden the scope of the problem, providing a more comprehensive perspective while also introducing entirely different frames for interpreting the problem at hand (Verganti 2009, 2016; Verganti and Shani 2016). According to Verganti (2009), external experts can play the role of "interpreters," stepping back from the current frame adopted by the firm and facilitating a more holistic interpretation of the surrounding context. Thus, problem reframing can result from heterodox interpretations and envisioning (Verganti and Öberg 2013). Although the notion of external input is not new, especially in the case of radical innovation (e.g., Capaldo 2007; O'Connor and Ayers 2005), the contribution of external experts, particularly to reframing activities, and how it unfolds has been only marginally explored (Fisher et al. 2018). In this study, we focus on the role of external experts prior to idea generation, namely, how they contribute to reframing the innovation problem.

To this end, we draw on the theoretical lens of reasoning logics—systems of thought used to draw inferences or reach conclusions based on premises, as originally proposed by Peirce (1998). Reasoning logics such as induction, deduction, and abduction shape how individuals frame and reframe problems (Cornelissen and Werner 2014), as they structure the way people make sense of information and generate new ways of thinking about the situation (Stierand and Dörfler 2016; Ulrich et al. 2015; Verganti et al. 2021). Reframing is based on interpretive processes through which individuals construct new meanings that guide action (Maitlis and Christianson 2014; Weick et al. 2005). These processes are closely intertwined with the enactment of reasoning logics in practice (Mumford et al. 1994; Baer et al. 2013). Dialogue is a key vehicle for this unfolding: it is a mutual process of verbal exchange through which individuals co-construct meaning, resolve ambiguity,

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

and articulate reasoning logics in interaction (Isaacs 1999; Tsoukas 2009). As such, dialogue enables people to either reinforce existing frames or creatively reframe them into new ones.

Fischer (2001) notes that all reasoning logics—deduction, induction, and abduction—aim to uncover new insights and expand our understanding of the world, each with different capacities to generate new knowledge. Therefore, by drawing on the theoretical lens of reasoning logics (i.e., Dong et al. 2016), we address the questions: *How do external experts apply reasoning logics to support firms reframe innovation problems?*

We address our research question through an ethnographic study (Fetterman 1998) in which we observe the involvement of experts in the strategic innovation projects of four global corporations. After observing the interactions and dialogues with external experts, we propose a process model that explores how problem reframing occurs, building on theories of deductive, inductive, and abductive reasoning. By coding the thought sequences and identifying the components of abductive thinking, classifying them, and examining their interrelations with deductive and inductive thinking and outcome variables (Garbuio and Lin 2021), this research setting allows us to extend current knowledge about the reframing of innovation problems.

Our study makes several contributions to two distinct literature streams: the reframing of innovation problems (e.g., Reynolds et al. 2023; Magistretti et al. 2025) and reasoning logics (e.g., Cramer-Petersen et al. 2019; Garbuio and Lin 2021). First, we show that external experts can contribute significantly to problem reframing not only by providing new knowledge within a problem space, but also by reframing the way a team makes sense of existing knowledge and arrives at a new meaningful problem. Second, we describe the interplay of five different reasoning logics in problem reframing: reverse deduction, reverse induction, explanatory abduction, speculative and moral innovative abduction. Specifically, we find that deductive and inductive reasoning can be crucial for defining the conversation space. External experts engage in the process through reverse deduction and inductive reasoning, attempting to create a common new frame, but their diversity forces them to seek new interpretations through abduction. Our study also identifies two different types of innovative abduction that can support reframing activities: speculative and moral. While speculative innovative abduction aims at imagining unconventional or paradoxical hypotheses that open up unexpected and breakthrough possibilities, moral innovative abduction relies on values that experts perceive as good and meaningful for the world. At a broader level, we envision this study as contributing to research on external stakeholder engagement by focusing on the role of external experts (e.g., Sukhov et al. 2021). Our findings can also inform studies that examine other types of external stakeholders, such as consumers (e.g., Gemser et al. 2025) or suppliers (e.g., Patrucco et al. 2024). More generally, the study contributes to the literature on the front end of innovation (e.g., Eling and Herstatt 2017; Wyrtki et al. 2021), aligning with research that seeks to provide guidelines for navigating this phase, which is widely recognized as the most uncertain and human-driven stage of the innovation process. Beyond the theoretical implications, our study also provides valuable insights for managers dealing with complex and ambiguous innovation problems, namely the potential contributions of external experts in reframing activities and the type of dialogue that can be established with them.

2 | Theoretical Background

2.1 | Reframing Innovation Problems

As Cornelissen and Werner (2014) point out, the framing construct has broad appeal in management and organization theory. Beginning with Burke's (1937) seminal formulation, it has since been applied in diverse fields ranging from linguistics and linguistic anthropology (e.g., Hymes 1974) to sociology and social movement research (e.g., Snow and Benford 1988), cognitive psychology and behavioral economics (e.g., Kahneman and Tversky 1979), journalism and mass communication research (e.g., Scheufele 1999), and design (e.g., Dorst 2011). Holding a managerial perspective, framing deals with the deployment of frames—knowledge structures that help individuals to interpret an incoming new reality—to select some aspects of the perceived reality and make them more salient (Giorgi 2017). In the field of innovation management, the identification of innovation opportunities, in the early stages of an innovation process, is greatly enhanced by the ability not only to frame, but also reframe the contextual factors that affect the innovation problem (Kim and Mauborgne 2004; Seidel 2007). Studies in design (Dorst 2015) and innovation (Reynolds et al. 2023), emphasize the importance of reframing the problem over ideation, meaning framing the innovation problem in an alternative and more meaningful way (Reynolds et al. 2023; Bianchi and Verganti 2021). In a world where ideas and solutions are abundant, reframing problems in novel ways is essential (Verganti et al. 2021).

Innovation problems are often characterized by the lack of a clear path to a solution because they do not specify the problem-solving goals, information, assumptions, or boundaries (Abdulla et al. 2020). They are ambiguous and include conflicting assumptions and information that can lead to different solutions (Reiter-Palmon 2017). If they are not framed in a sufficiently defined form prior to the solution effort, fewer original outcomes will be achieved (Silk et al. 2021). Although problem reframing is recognized as fundamental to fostering innovation through creativity (Abdulla et al. 2020; Magistretti et al. 2025; Unsworth 2001), innovation studies have mainly focused on the generation of solutions (Amabile 1988; Kaplan and Vakili 2015) or their evaluation (Berg 2019; Mueller et al. 2014).

Recent research calling for the investigation of how innovation problems are framed and reframed, and not just how they are solved, has led to a resurgence of interest in problem reframing in the innovation community (Posen et al. 2018). Valuable insights in this direction have emerged from studies in the field of design. As Bianchi and Verganti et al. (2021) highlight, the design discipline offers a rich and insightful body of literature on problem reframing. Dorst (2011) has contributed extensively to the problem reframing debate, identifying the ability to frame a problematic situation in new and interesting ways as a key characteristic of design. Drawing on Schön's (1983) view of design as a reflective practice, several design scholars have contributed to the sensemaking perspective (Stompff et al. 2016;

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Van Leeuwen et al. 2020), whereby designers frame and reframe the problem in their minds through subjective lenses. Problem reframing requires intense engagement in sensemaking to develop a deep understanding of the problem, as well as dialogue involving inquiry, divergence, and convergence (Fairhurst 2005; Senge 1991). Inquiry forces individuals to recognize and then challenge the assumptions, acknowledging the limitations of their own perspectives (Beckman and Barry 2009; Schein 1999). Critical thinking underpins the reframing activities (Gottschall 2013; Paul and Elder 2013).

In practice, reframing occurs through reasoning, which enables individuals to generate new ways of thinking about a situation. Such reasoning often involves comparing familiar frames with alternative ones that cannot be fully grasped or articulated; this comparison allows reasoning to trigger reframing by introducing new frames and inferences (Cornelissen and Werner 2014). Reasoning is enacted through direct interaction in the form of dialogue and gesture. Engaging in reframing through dialogue and interaction allows for the creation of new frames to make sense of the situation in novel ways (Bellis, Magnanini, et al. 2024). Novel frames emerge as assumptions, or thoughts accepted as true without proof, typically based on incomplete and inconsistent knowledge (Peirce 1998). These assumptions are evaluated according to their relevance and plausibility (Garbuio and Lin 2021). Relevance refers to how well an assumption fits the new scenario and is contextual and temporal, shaped by the current state of knowledge (Magnani 2016). Plausibility refers to the extent to which an assumption provides a probable explanation of *what is* and *what ought to be* (Weick 1995). Therefore, reasoning logics represent an appropriate theoretical lens to explore how problem reframing unfolds. In the following sections, after introducing the role of external experts in reframing, we further expand the discourse on reasoning logics and their role in reframing.

2.2 | Role of External Experts in Reframing

Several studies underline the relevant contributions of external experts to early stages of innovation, some focusing on ideation (e.g., Chesbrough 2003; Chesbrough and Bogers 2014) or idea evaluation (e.g., Sukhov et al. 2021), but rarely investigating their contribution to problem reframing (e.g., Jepsen et al. 2014; Verganti 2009). The sensemaking endeavor inherent in reframing requires drawing on past experiences to reinterpret existing understandings in novel ways (Maitlis and Christianson 2014). Involving experts in problem reframing allows the problem to be viewed from different perspectives, unencumbered by organizational frames (Ainamo et al. 2021; Altuna et al. 2017; Verganti and Shani 2016). While organizations tend to frame and interpret reality based on their stories and experiences (Altuna et al. 2017; Cukier et al. 2021), this can be limiting or misleading because it does not provide a comprehensive view of the problem space (Bianchi and Verganti 2021). Experts provide new lenses to make sense of the situation in novel ways, shifting the focus to alternative aspects that were not part of the original problem space but were then creatively brought in (Fisher et al. 2018; Dorst 2015). Thus, the experts who intervene in problem reframing are not necessarily specialists in the product or service that the firm is trying to innovate, but they are thought leaders, pioneers, or professionals in the area under study. In particular, they all look at the same phenomenon, at the same "person" (client, consumer, category of people, etc.) from different perspectives. Verganti (2009, 2017) speaks of "interpreters" who are interested in the same "experience of life" but look at it from different perspectives and are therefore able to reframe the firm's understanding toward a more comprehensive view of the problem space.

In the sensemaking perspective, Sukhov et al. (2021) explain how experts can both generate new ideas and reframe existing understandings in new ways by reducing ambiguity and enabling future action. Given the sensemaking nature of problem reframing activities, scholars agree on the participatory nature of expert involvement: it is not only about knowledge transfer (Nonaka and Takeuchi 1995) but also especially about knowledge creation through interactions between different actors (Sukhov et al. 2018). To some extent, firms engage experts for problem reframing not to compare overlapping issues and expertise but to integrate and juxtapose different points of view and frames to build the big picture. Therefore, understanding how external experts contribute to the reframing of innovation problems could shed light on the process by which these experts reason alongside firms in the innovation reframing process.

2.3 | Reasoning Logics in Reframing

To explore how external experts contribute to problem reframing, we adopt the theoretical lens of reasoning logics. Early studies about reasoning logics were conducted by Charles Sanders Peirce (1998), one of the founders of Pragmatist Philosophy, who devoted many years of his life to the study of deduction and induction, models of thought that he supplemented with another logical inference, abduction. As previously reported in this paper, scholars emphasize that problem reframing essentially relies on reasoning logics (e.g., Mumford et al. 1994). As Fischer (2001) notes, according to Peirce, all reasoning logics have the goal of discovering something new and expanding our knowledge of the world. However, the philosopher also affirmed that three specific logical inferences have a different quality in terms of their ability to create new knowledge: deduction, induction, and abduction. Although several authors highlight the fundamental role of abduction in fostering innovation (Chiffi and Pietarinen 2020; Cunningham 1998), few studies explore the interdependencies of all three reasoning logics in addressing innovation problems (Cramer-Petersen et al. 2019; Dong et al. 2016; Eisenmann et al. 2012; Garbuio and Lin 2021).

In the philosophy of science and logic, *deduction* is considered truth-conserving, that is, conclusions are necessarily true if the premises are true and implicit in the premises. It proceeds from the general (the rule) to the particular (the result). Deductive reasoning can be defined as self-referencing because reaching a conclusion is based on the logical implications of two or more propositions asserted as true (Magnani 1995; March 1976; Reichertz 2010). Deduction refers to the application of general rules to specific cases (Richardson and Kramer 2006; Schurz 2008). As Folger and Stein (2017) note, deductive inferences have logical validity provided by their internal structure, and hence do not require further testing or proof. In the case of

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

induction, the premises are empirical results and observations. According to Fischer (2001, 366), "the inferred conclusion is considered to be content-increasing, but not truth-conserving because the inference is only an assumption that cannot be proved with ultimate certainty." Unlike deduction, induction moves from the particular to the general. Johnson-Laird (2008) argues that inductive reasoning is based on the process of deriving plausible conclusions that go beyond the information in the premises. Similar to deductive reasoning, inductive reasoning is self-referential, in that it infers concepts only from available data in a model or frame of reference (Magnani 1995; Reichertz 2010; Schurz 2008).

According to Peirce, *abduction* can be defined as "the process of forming an explanatory hypothesis" that fosters creativity and innovative thinking by allowing decision-makers to adopt a logic of what might be. The abductive process allows the creation of new order parameters after a surprising event and new habits stimulated by creativity (Gonzalez and Haselager 2005). While induction is based on hypotheses extrapolation, abduction allows imagining new hypotheses. Chiffi and Pietarinen (2020) argue that abduction can be used to anticipate future scenarios and make sense of ambiguous information, hence relevant to exploring how this reasoning can support strategic thinking, help managers look for something new, and make strategic decisions. According to Cunningham (1998), abduction is the appropriate reasoning to make sense of new or unknown situations. Dunne and Dougherty (2016) emphasize that abductive reasoning helps innovators detect useful information (among noise and competency traps) and holistically accumulate insights. Roozenburg (1993) and Dorst (2011) distinguish two types of abduction: (i) explanatory abduction, which aims to create something that operates with a known working principle in a given value creation scenario; (ii) innovative abduction, which aims to create something and the working principle, knowing only the desired end value. Garbuio and Lin (2021) distinguish the role of explanatory and innovative abduction in problem finding according to the level of definition of the solution. When the problem is unknown, but the solution is defined, explanatory abduction can support the identification of a possible explanation (i.e., hypothesis) from an observation and a rule that explains the observation. When both the problem and the solution are unknown, innovative abduction is more fruitful in explaining an observation (unexpected outcome or desired/predicted value) by first introducing an innovative idea (value to be created) and then the conditions (how to solve the problem).

3 | Methodology

To address our research question and investigate how external experts apply reasoning logics to help firms reframe innovation problems, we conducted an ethnographic study that allowed us to directly observe and analyze the reasoning logics of the external experts involved in the strategic innovation projects. Our study focuses on four global firms supported by Alpha (anonymized), an international innovation consultancy that supports firms in their strategic innovation projects and relies on external experts for problem reframing activities. In view of the exploratory nature of our study, we considered the ethnographic approach to be the most appropriate, as it is a discovery-oriented research practice with a long tradition of providing insights into new or unappreciated processes that are important for understanding how work is done (Locke 2001). Given our interest in how experts reason and how their reasoning is combined in problem reframing activities, the ethnographic approach was preferred to others (e.g., case studies). This would not have been possible with interviews, for example, which are retrospective in nature. Ethnography, by contrast, involves prolonged immersion in the organizational contexts and enables a richer, more comprehensive observation of direct interaction and dialogues, allowing researchers to capture reasoning as it emerges in experts' words. Rather than relying on retrospective accounts, our data were generated primarily through participant observation, informal interactions, and detailed field notes capturing how experts and internal teams engaged among them, exchanged interpretations through dialogue, and iteratively contribute to problem reframing in real time. This stance positions our focus not on the four projects as discrete "cases," but on the situated interactions through which reasoning logics are enacted. Therefore, an ethnographic approach allows us to observe reasoning logics as they occur, to trace how meanings and frames evolve across interactions, to follow the trajectory of conversations and problem reframing over time.

3.1 | Research Setting

Alpha is a consultancy focused on innovation management, supporting firms in developing new product concepts through problem reframing, opportunity identification, and vision (Eling and Herstatt 2017) by involving external experts. For each project, Alpha forms small teams (4–6 consultants) who work closely with firms. As ethnographers, we observed four Alpha projects between February 2020 and May 2022. Due to COVID-19, all were conducted remotely via digital platforms like Zoom and Microsoft Teams, and collaborative tools such as Miro or Mural. This setup enabled detailed observation and improved data comparability across projects. Besides, it leveled the playing field from the participants' perspective, creating a neutral space for interaction. However, we acknowledge its limitations in that it inhibits real interaction and hinders communication based on gestures and gut feelings. Table 1 provides an overview of the four projects.

Alpha applies a consistent, consolidated methodology across all four projects, each lasting 5 to 8 months depending on contextual factors. Consultants take on dual roles as advisors and coaches throughout the innovation journey, attending closely to organizational culture, values, and practices.

Each project is structured in three main steps: *early opportunity identification and preliminary problem framing, external expert engagement*, and *preliminary concept design*. Although only *external expert engagement* is of interest in our study, we had the opportunity to follow and observe the entire journey in all four projects. Specifically, we were able to participate as observers in all activities, with soft involvement in the facilitation activities during workshop moments (i.e., guiding participants through the different activities and providing support with the digital tools adopted when needed). Our involvement throughout the journey was critical to understanding the antecedent dynamics

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 | Projects overview: Aim and description.

Project ID	Project aim	Industry	Core business	Category	Project team location
Project 1	Identify emerging concepts for skincare products specifically focusing on Gen Z	Skincare	Skincare products (creams, serums, facial treatments, etc.)	Large Multinational Corporation	Japan
Project 2	Identify emerging concepts for haircare products specifically focusing on Gen Z	Haircare	Haircare products (shampoos, conditioners, masks, etc.)	Large Multinational Corporation	Singapore
Project 3	Identify emerging concepts in the pet industry, focusing on pet care products	Pet care	Pet care products	Large Multinational Corporation	United States
Project 4	Identify emerging concepts in the photography industry, focusing on accessories for professional photographers	Professional photography	Photography equipment	Large Corporation	Italy

and effects of the phase of interest (see the data collection section for details). In addition, it allowed us to closely observe how and why the agents acted, felt, and thought the way they did (Wacquant 2002). *Early opportunity identification* consists in engaging a team of about 20 people from the organization in a problem framing exercise. The team includes all those involved in defining the firm's strategy, as well as managers from key areas whose contributions may be crucial to identifying innovation opportunities. The output of this framing phase is a set of *hypothetical frames*, which are interpretations addressing an identified innovation opportunity (Cornelissen and Werner 2014). For example, in Project 2 (Haircare), the innovation opportunity identified was to bring "haircare closer to the idea of changing clothes." The underlying hypothetical frame was that "Gen Zers perceive their identity as fluid, temporary, and ever-changing." The main limitation of the hypothetical frames is that they are imbued with the knowledge and perspective of the specific organization. *External expert engagement* consists in involving external experts with the aim of better reframing the problem space and elaborating the identified *hypothetical frames* toward *renewed frames*. *Preliminary concept design* consists in translating the renewed frames into innovative ideas for new products. In the following, we focus on the *external expert engagement* phase.

The defined problem space and *hypothetical frames* were the result of collaborative efforts among key actors internal to the organization, and their quality needed to be assessed for relevance and plausibility (Garbuio and Lin 2021). Alpha and the client firm took several steps that culminated in a session with external experts, as described below (see the Appendix S1 for a detailed description). As a first step, Alpha and the client firm deconstructed the *hypothetical frames* and isolated the underlying *assumptions* (e.g., Cramer-Petersen et al. 2019). This activity is fundamental because the *hypothetical frames* are imbued with the firm's concepts, values, and culture. In contrast, external experts from different domains can easily validate, reshape, or reject the assumptions (see Appendix S1 for more details on the formulation of assumptions). As a second step, the firm and Alpha began identifying external experts to involve, with the primary goal of having an appropriate diversity of expertise in the client's context of use to cover all areas of the assumptions (Table 2 summarizes the characteristics of each external expert).

The final step was the external expert session, which lasted an average of 3 h, during which the experts and managers engaged in a dialogue to assess the plausibility and reliability of the assumptions. During the session, each external expert was given a 20-min time slot to answer questions. All workshop participants (company team and other experts) were invited to intervene and build on the external expert's comments with additional insights or questions. The session ended with everyone in the room (external experts, Alpha team, company team) writing down and sharing two critical insights that emerged. The session was moderated by a member of the Alpha team, which was crucial in terms of logistics (time management and ensuring enough space for all external experts), content, and questioning. In general, all the expert sessions were conducted in a relaxed manner to maximize participant interaction. The firm acted as the host during the external expert session but actively participated in the conversation with the external experts, offering the firm's perspective on the identified assumptions and the client's context of use.

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Experts involved in the four projects (for online publication only).

Project ID and industry	Experts engaged			
	Cluster	**Industry**	**Role**	**Country**
Project 1 (Skincare)	Adjacent	Food	Lead of concept and prototype design	Switzerland
	Adjacent	Medicine	Dermatologist	China
	Outside the industry	Gaming	Cultural foresight	United States
	Outside the industry	Fashion	Brand development	Italy
	Outside the industry	Music	General manager of artist management division	Japan
	Outside the industry	Consumer Insights	Chief operating officer	China
	Outside the industry	Fashion	Influencer	Japan
Project 2 (Haircare)	Adjacent	Personal care	R&D researcher	Japan
	Adjacent	Personal care	R&D researcher	Japan
	Outside the industry	Fashion	Retail excellence director	Italy
	Outside the industry	Fitness apparel	Digital marketing	Netherlands
	Outside the industry	Food and beverages	VP of marketing	United States
	Outside the industry	Education	Head of marketing	Germany
	Outside the industry	Gaming	Consumer insights	United States
	Outside the industry	Professional	Anthropologist	Sweden
	Outside the industry	Pharma	Digital strategies	Italy
Project 3 (Pet care)	Outside the industry	Hotelier	Senior director of room operations	Austria
	Outside the industry	Home design	Marketing manager	Sweden
	Outside the industry	Professional	Pet behaviorist	United States
	Same industry	Pet retailer	CEO of pet retail chain	United States
	Same industry	Pet products	Co-founder of pet product brand	United States
	Same industry	Pet products	CEO of pet food brand	New Zealand
	Adjacent	Pet friends	Pet sitter	United States
Project 4 (Professional photography)	Adjacent	Photography digital services	CEO of photography agency	Italy
	Same industry	Product photographer	Product ambassador	Israel
	Same industry	Wildlife photographer	Product ambassador	United States
	Outside the industry	Personal care	Creative director	Japan
	Same industry	Camera manufacturing	Product manager	Japan
	Adjacent	Education	University professor	United Kingdom
	Outside the industry	Eyewear	Content manager	Italy
	Adjacent	Photography digital services	Development manager	United States
	Adjacent	Photography digital services	Head of strategic development	United States
	Adjacent	Photo education	Educator	United Kingdom

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 | Data sources and use.

Data source	Type of data	Use in the analysis
Archival data	*Company website and social media pages*. *Project-related documents*: company material and presentations related to the state-of-the-art of the product area subject to innovation.	Familiarization with the organizational context, values, and language. Support the reconstruction of the set of concepts already produced in each project. Support, integrate, and triangulate evidence from observation and workshop recordings.
Project materials	*Project-related emails (around 1.000)*: emails containing content used within the project (not considering quick email exchanges to arrange meetings). *Project workshop material (187 documents)*: PowerPoint presentations used to guide the workshops' sessions, PowerPoint presentations and other documents produced by participants in the workshops, textual artifacts, drawings, or text boxes produced on Miro.	Support, integrate, and triangulate interpretations from the workshop sessions. Keep a record of the project outcomes that emerged and share with the second and third authors. Familiarization with the organizational context. Integrating observations to improve our understanding of project-related decisions. Connect specific insights and outputs of different project phases. Identify hypothetical frames, renewed frames, and related assumptions developed by the firm.
Observation	*Field notes from attending meetings (146 meetings)*: detailed record of conversations and the evolution of the innovation direction from early stages until the final presentations. *Informal conversations*: informal talk with managers before and after meetings.	Familiarization with the organizational context, gain the trust of informants, discuss insights from observation, clarify uncertainties about project-related decisions, and support emerging interpretations. Keep a record of the outcome of practices that members engaged in during the projects (e.g., specific jargon, sketching) and share with the other authors.
Recordings	Experts' preliminary online interviews (35 interviews, 2100 min of recordings): interviews to capture experts' perspectives of the assumptions to be tested. External expert sessions (4 sessions, 780 min of recordings): recordings of the workshop sessions where the firms met the external experts with the aim of assessing the proposed product concepts through the interpretation of the assumptions.	Integrate observations with informants' accounts to improve our understanding of the social dynamics and project-related decisions. Extract the specific reasoning logics used by experts when asked to interpret a specific assumption or when interacting with others. Investigate how experts engaged in the conversation and how the reasoning logics led to a renewed frame.

The setting described was optimal for observing the reasoning of different experts. The consultancy created a space for dialogue, but did not drive the conversation. While the experts were free to reason, interact with each other, and with the firm's team, the reasoning logics and their sequencing emerged directly by the individual experts; they were not predetermined or guided by the consultancy's specific collaboration.

3.2 | Data Collection

Data collection followed the general recommendations for ethnographic research (Fetterman 2019; LeCompte and Schensul 1999; van Maanen 2011) and combined archival search, participant observation, notes from direct participation in meetings between Alpha and the client firms to design the activities, project materials and reports, and informal conversations. Table 3 describes our data sources. Across the four projects, the first author attended all the meetings and workshop sessions from the initial kick-off meeting and project design to the closing session. These meetings, in addition to the workshop sessions, were crucial in familiarizing herself with the firms' business, the organizational context, and language. During these meetings, she took notes on the project content and the dynamics among participants to capture as much of the conversation verbatim as possible. She then rewrote and expanded these notes after the meeting while her memory was still fresh (Emerson et al. 1995). Most meetings involved the entire Alpha team and one or two members of the client team, including the project manager.

Since all projects were conducted virtually, all the external expert sessions and interviews were recorded to facilitate ethnographic observation. During the project activities, the authors focused on the development of the project, knowing that they could later review all the activities and interactions from the recording. In addition, this allowed the authors who did not participate in the observation to review them with fresh and unbiased eyes. In total, we collected 2880 min of recordings, which were also specifically requested by the firms for later internal use (e.g., allowing other people across the organization to view the experts' contribution). During the development of

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

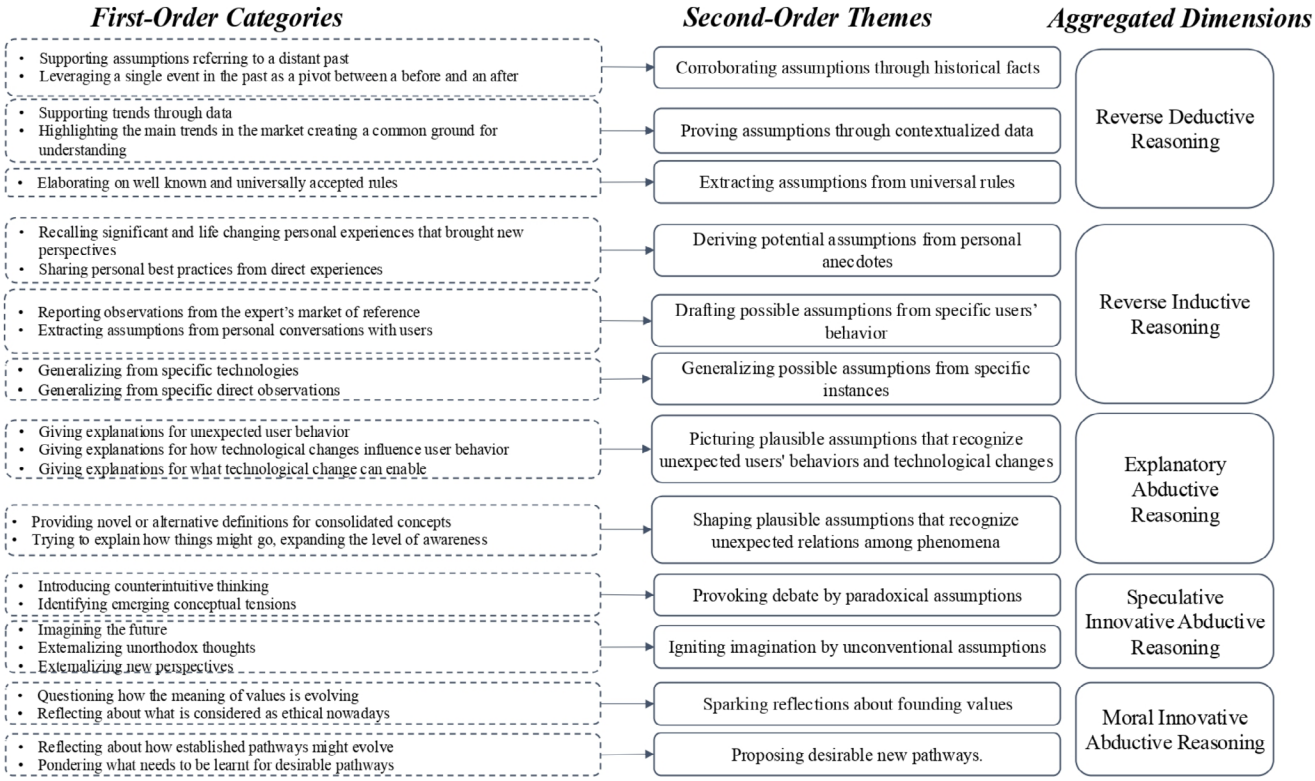

FIGURE 1 | Coding tree.

the projects, a large amount of material was produced in the form of PowerPoint presentations, emails, and other documents to record the outcome and share it with the other authors. In addition, most of the collaborative activities during the project session were conducted using Miro, a brainstorming and project management platform that allows people to collaborate using images, drawings, post-it notes, or text boxes. All material and textual artifacts produced during the sessions (boards, sketches, etc.) were saved to keep a record of the practices people engaged in during the project.

3.3 | Data Analysis

Consistent with our research goals, our data analysis focused primarily on the information gathered during the *external expert engagement*, using the data (recordings, notes, and transcripts) to analyze the emergent reasoning logics and interaction dynamics. The transcripts of the interviews and external expert sessions constituted our primary data to extract the experts' reasoning logics and to analyze how they related to each other. We also used the field notes and session material to support and refine the interpretation of the emerging categories and guide their integration into an overall framework. Finally, data collected during the *early opportunity identification* and *preliminary concept design* phases were critical to delineating the hypothetical frames, their underlying assumptions, the renewed frames, and their associated assumptions.

As a first step, the first author open-coded the interviews and recordings of the external expert sessions, searching for relevant text segments—phrases and passages indicating the interpretation of the assumptions through recognizable inferential logics. The first author initially labeled these segments with the in vivo terms and phrases used by the external experts (Locke 2001), generating a large dataset of codes. After multiple re-readings of the data, she gradually combined the in vivo codes relating to similar dynamics or behaviors, albeit varying in specifics, into first-order categories (Locke 2001). The other authors then reviewed the coding and together resolved any discrepancies by discussing and recoding the data as needed, comparing the first-order categories and structuring them into second-order themes (Gioia et al. 2013). We labeled the second-order themes with metaphorical expressions alluding to how external experts joined the conversation and their reasoning (e.g., "corroborating assumptions through historical facts," "deriving potential assumptions from personal anecdotes," "provoking debate through paradoxical assumptions"). In addition, we used axial coding to examine and compare the second-order themes with the main reasoning logics—deductive, inductive, explanatory abductive, innovative abductive—to formulate theoretically informed aggregate dimensions (Gioia et al. 2013; Van Maanen 1979). We verified internal validity through pattern matching. Subsequently, an external researcher validated the codebook to increase the reliability of the research process. Figure 1 shows the coding tree. Looking at our codebook at the aggregated level, we noticed how the less adopted reasoning logic seems "reverse deductive reasoning," while "reverse inductive reasoning" emerges as the most used. While, in terms of abduction, external experts relied more on "innovative abduction" than "explanatory abduction." Among the two different innovative abduction identified in our data analysis, "speculative innovative abduction" emerges more clearly than "moral innovative abduction." Finally, we examined the links between our

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

aggregate dimensions to build a coherent model that explains how experts use reasoning logics to reframe a hypothetical frame into a renewed frame.

4 | Findings

In exploring how external experts apply reasoning logics to assist firms reframe innovation problems, we observed two key aspects. First, how the innovation problem is reframed, namely, how the hypothetical frame developed by the firm is transformed into a renewed one. Second, how the external experts perform the reasoning logics to assist firms in reframing processes.

4.1 | Problem Reframing: How Hypothetical Frames Evolve Into Renewed Frames

By observing how hypothetical frames are transformed into renewed frames, our data clearly show the difference between the two. For example, in Project 2 (Haircare), one of the hypothetical frames developed by the firm relates to the identity of Gen Zers, which was reframed as shown in Table 4. The reframing process reveals a significant shift: while the hypothetical frame defines identity as *fluid and temporary*, the renewed frame adopts a "both and" perspective. This redefinition offers an entirely different interpretation of identity, not as something predefined that Gen Zers must adapt to (even if constantly changing), but as something they can actively create and experiment with according to their preferences. Similarly, in Project 3 (Pet Care), one of the hypothetical frames developed by the firm initially framed the human-pet relationship as *sharing time and space*. However, the renewed frame provided a deeper perspective, redefining this relationship as a *lifelong bond in which the pet grows alongside the human*.

The expert' reasoning logics played a crucial role in the transition from the hypothetical to the renewed frame. As outlined in the methodological section, the hypothetical frame was communicated to the experts as a set of assumptions. These assumptions acted as triggers for the experts' reasoning, leading to new insights and the formulation of new assumptions, and thus a renewed frame. Our analysis of the experts' reasoning revealed the use of all the main logics—deductive, inductive, exploratory abductive, and innovative abductive—and how they collectively contributed to transforming a hypothetical frame into a renewed one, thereby enabling problem reframing. In Appendix S1, we provide two examples (Figures A.1 and A.2 in Supporting Information) that illustrate how the experts' reasoning (captured in their own words) about the firm's hypothetical frame emerged and facilitated the development of a renewed frame, thereby reframing the problem. Precisely, we observed that each external expert started with a preliminary elaboration of the assumptions derived from the hypothetical frame, through reverse deductive and inductive reasoning. Later, they engaged in a generative phase through explanatory and innovative abduction, which led to the development of insights. These insights naturally embedded in the subsequent experts' contributions and, cumulatively, in the development of the renewed frame. Below, we present how each reasoning logic is performed by external experts and discuss its implications for problem reframing.

4.2 | Reverse Deductive Reasoning (What It Must Be)

Deduction is a form of logical reasoning that establishes the validity of the conclusion if the premise of the rule is found to be true (Dong et al. 2016). Given the evaluative nature of the process under study, we observed that experts relied on a type of *reverse deductive reasoning*, that is, instead of starting from a premise to support a conclusion, they started from the conclusion (the assumption to be evaluated) and searched for the best premise to support that conclusion. For example, in Project 3 (Pet care), one of the assumptions was that the human-pet relationship is a daily bond and that separation is difficult. An external expert justified this conclusion with the following premise:

> Because of all the restrictions with COVID 19, the airlines have limited the number of pets that can be brought into the cabin, but also the overall number of pets that can brought onto the plane. So there is a lot of change in the mindset of our travelers right now

TABLE 4 | Reframing innovation problems: Examples of hypothetical and renewed frames.

	Hypothetical frame	Renewed frame
Project 2[a]	*Statement*: Gen Zers' identity is dynamic fluid, and temporary.	*Statement*: They are the 'BOTH AND' generation (non-binary). Why choose when you can be both?
	Description: Gen Zers will hardly look for a strong identity and assert this identity. They don't aspire to fit anymore. Who they are is connected to how they feel, and this may change over time.	*Description*: By living in the moment and constantly changing and experimenting, they experiment with their carefully curated identities.
Project 3[a]	*Statement*: People want a pet not to own it, but to build a relationship.	*Statement*: A pet as a personal companion.
	Description: Having a pet is about enjoying time and space together.	*Description*: The pet as a human companion to grow with.

[a]See the Appendix S1 for details on reframing the innovation problem.

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

> because they are not able to take their pets anywhere whenever they want.

In our data, we observed that the experts relied on different premises. On the one hand, they used historical facts to distance themselves from the past and evaluate the assumptions to show how the world has changed. This detachment from the past occurred mainly in two ways. First, referring to the distant past and showing how the world is different today, as reported in a quote from Project 4 (Professional photography):

> 30 years ago, probably a bit longer, the only way to get [a photo] out there was to kind of get it published in magazines or put it on, uh, a gallery wall. And now, you know, everybody's their own creator and distributor and I just put my pictures on Instagram and I can send them to whoever I want around the world.

Second, by identifying a pivotal event that allowed a "before" and an "after" to be clearly identified, as noted in a quote from Project 3 (Pet care):

> I mean, Amazon and Alibaba were already there before Covid 19, but Covid 19 really accelerated these, and this brings big, big changes in the way in which companies must be managed as well.

On the other hand, inverse deductive reasoning also relied on contextualized data to support and consolidate the existence of certain trends, as in a quote from Project 4 (Professional photography):

> And this is a trend of course, nowadays in our community, the well-known brands still excel. I took some numbers just to give an idea: 40% own a Brand 1, 30% a Brand 2, and 11% a Brand 3 camera. That's interesting. Because these are the great changes that we are seeing.

Alternatively, contextualized data could highlight key market trends or create common ground for further discussion, as a quote from Project 2 (Haircare) highlights:

> Gen Z in China are very different from Gen Z in Japan and Gen Z in Europe, or the western world. And, I think that the main reason is because of the culture or the big political movements happening in China.

Indeed, the press coverage of major political movements in China, emphasizing these as a crucial element in defining the problem, seems to support the argument that people are bound by something objectively recognizable. Finally, we observed reverse deductive reasoning referring to universally accepted rules. This is the purest form of deductive logic, where the analysis is developed through the use well-known and shared rules. For example, an expert referring to photography (Project 4—Professional photography) stated:

> It is clearly an artistic expression. Okay? So as any artistic expression, it tries to create an emotional connection with the viewer.

In this quote, the universal rule is in the fact that artistic creations enable emotional connections, the basic goal of any artistic production. This reasoning logic does not create new knowledge or insights but seeks to confirm and reinforce existing beliefs: experts look for *what it must be* in assessing the assumptions. Reverse deductive reasoning appears to be crucial in setting the boundaries of the conversation for problem reframing and creating anchors for the assumptions. Even without thinking deeply about the assumptions, the experts delineated the boundaries of their own mental models and the high-level beliefs that drive their understanding of the assumptions in the preliminary evaluation (see Table 5 and Appendix S1 for selected quotes).

4.3 | Reverse Inductive Reasoning (What It Is)

Induction is a form of logical reasoning in which rules are developed based on some observations, allowing for generalization and empirically based conclusions (Dong et al. 2016). Similar to inverse deductive reasoning, we observed a form of reverse inductive reasoning, where instead of inferring conclusions from empirical results and observation, the experts started from the conclusion and linked it to the most appropriate recognized pattern. Using the same example of the human-pet relationship, an external expert looked for confirmation of this conclusion in the following premise, delineating a pattern:

> I had several situations where I had to make the decision to leave my pet with my parents. When I came back, I immediately felt my pet connected to me.

Even with reverse induction, the patterns and observations came from different sources. When joining the conversation, the experts sometimes started with themselves, providing personal experiences and anecdotes directly linked to the conclusion, as reported in the quote above. We observed that the sharing of personal experiences served several purposes. For example, experts referred to significant and life-changing experiences that brought new perspectives, as reported in a quote from Project 4 (Professional photography):

> We know the system quite well because the first photographers were me and Jack, my co-founder. So, we kind of experienced this problem first hand. We understood that the key for the future is trying to save time for photographers to enable them to have time to improve their abilities, their vision. And also, in the most standardized system.

In addition, generating rules from observations based on personal anecdotes can also highlight best practices. For example, an expert evaluating an assumption about the relationship with pets (Project 3—Pet care) stated:

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 5 | Reverse deductive reasoning, selected quotes.

Aggregate dimension: reverse deductive reasoning	
Second-order themes	**Selected evidence of first-order categories**
Corroborating assumptions through historical facts	*Supporting assumptions referring to a distant past*
	I think in the 80s and 90s, we were all a little bit trained, brainwashed, and there was this sense of beauty, you know. Even if you take a magazine from those years and look at the ads, you see everything is fake, and everything is photoshopped and polished. I think that is completely over. And now we are in the era of more reality. So, you see people hanging out at home, laying on a sofa, but maybe without makeup, some more reality.
	Leveraging a single event in the past as a pivot between a before and an after
	I would also say that COVID-19 has actually put this trend into hyper drive.
Proving assumptions with contextualized data	*Supporting trends through data*
	We did a study on creativity, pretty recently, and it turned out that 74% of creatives say they spend over 50% of their time on repetitive, non-creative tasks.
	Highlighting the main trends in the market creating common ground for understanding
	Gen-z are unique because they have no rubric. Everything that the previous generation could refer to as a role model or the rules of how to survive this world does not work for them because many are quite radically denied. The global economy is proving that even well-developed industries also can be crushed, the economic system we developed is facing a difficulty. Many of the previous rules of success are denied.
Extracting assumptions from universal rules	*Elaborating on well-known and universally accepted rules*
	Key in trusting the brand is whether people are using the brand.

> I switched jobs a couple times actually during the pandemic, and I was looking for full-time work from home for one reason, this pet sitting business that I have. Because I saw the value in having a full-time job from home for these customers, because a lot of people want someone to be around.

Reverse inductive reasoning also emerged in the experts' observations of user behavior. All our projects dealt with products that have a specific target user, and all experts dealt with and drew their conclusions from observing the target user in his or her reference market, as reported in a quote from Project 4 (Professional photography):

> We're also seeing in the professional market more and more professional photographers (...) experimenting and exploring with video because there is so much more content now that is required for video because of the internet.

The experts also derived rules from their direct conversations with target users:

> Photographers, from the discussions and the talks we are having with them, they don't take many pictures actually, when they think about the project, you know, they engineer it before they go there.

At times, the experts relied on reverse inductive reasoning and specific instances they dealt with in their daily lives, such as technologies that are not widely used but in which they are experts and can therefore draw conclusions from recurring observations. For example, an expert in the fashion industry from Project 1 (Skincare) stated:

> Now with apps, you can scan the barcode, and they tell you in a rating from one to 10 how this cream is rated compared to sustainability and also compared to impact on your skin. And they give you on the app all the alternatives of what you can buy instead of this. So of course, it's also a marketing tool.

Similarly, they may generalize rules based on observations specific to their industry, as in a quote from Project 2 (Haircare):

> The first point is for sure they are not loyal. So, I'm speaking now from a fashion point of view, but I think it's general.

Regarding reverse induction in general, interesting to note is that the external experts did not start directly from consolidated knowledge or competencies related to the assumption to be evaluated, but from intimate and personal experiences to help others empathize with their perspective and to better receive and digest their personal view, almost as if starting from personal experiences helps the external experts make their perspectives more tangible. In both cases, concrete examples help to understand the implications of certain situations or actions, thus facilitating the problem reframing activity. Through reverse inductive reasoning, external experts introduced an additional level of uncertainty into problem reframing, as they did not rely on consolidated and validated knowledge, but on patterns of evidence, trying to explain *what it is* (see Table 6 and Appendix S1 for selected quotes).

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 6 | Reverse inductive reasoning, selected quotes.

Aggregate dimension: reverse inductive reasoning	
Second-order themes	**Selected evidence of first-order categories**
Deriving potential assumptions from personal anecdotes	*Recalling significant and life-changing personal experiences that brought new perspectives*
	I will tell you a little of my personal side, I went through a personal evolution with Bubu. Because of my experience, I started seeing things in a different way.
	Sharing personal best practices from direct experiences
	I always check the photographer website before selecting them.
Drafting possible assumptions from specific user behavior	*Reporting observations from the expert's market of reference*
	People are looking for greater freedom with their gear. People are getting lazy or spoiled. They're looking for smaller cameras, smaller lenses, smaller everything. Why, why do people carry around the smartphone and not a proper camera, even if the camera is better? This is easier. This is always with me. I don't need to carry a special bag or special equipment. It's only 100 g lighter, but people will spend a lot of money on lighter and smaller.
	Extracting assumptions from personal conversations with users
	I spoke to an art director about a magazine on a remote way to shoot photos.
Generalizing possible assumptions from specific instances	*Generalizing from specific technologies*
	Because of the advancement of the computer and AI
	Generalizing from specific direct observations
	Today it is very easy to become a photographer. I see so many young photographers.

4.4 | Explanatory Abductive Reasoning (What It Might Be)

Abductive reasoning introduces even more uncertainty than inductive reasoning because it brings in new theories and patterns to explain the assumptions (Dong et al. 2016). In explanatory abduction, people tend to infer rules that are known in other contexts but are new to the current context (Dong et al. 2016). In our data, unexpected observations and events emerged with respect to different elements. On the one hand, experts envisioned new assumptions as a result of surprising user behaviors and/or technological changes observed in the context of reference. Specifically, experts pointed to specific user behaviors that did not resonate with their beliefs and frames of reference to provide a plausible explanation for such behaviors, as reported in a quote from Project 1 (Skincare):

> I don't think they want to be different and unique so that they can accept themselves. I think the competition is even fiercer for the new generations. So, they want to be outstanding, no matter in what part. So, they have very strict criteria for themselves.

They also inferred new assumptions to provide plausible explanations for what the technological changes may enable. An example from Project 4 (Professional photography):

> If you're talking about the commercial sphere, e-commerce is now possibly the major form of commerce and retail. Therefore, instead of having a physical environment that you walk into in and browse, you have a digital environment that you've become accustomed to browsing within. And within that you go and look for things. If I go and look for a toy for my grandson, and there's a static image of it on a white background, which is very much what Amazon does, and you look at them, then a lot of the intent to buy comes from the fact that you trust the brand, that Amazon is a trustworthy brand and that what you're looking at is what you're going to get. Besides, there's a description. So, the photograph becomes a stimulus, and then there's a whole layer of data around it.

Finally, experts generated new assumptions to interpret how technological changes affect user behaviors in unexpected and surprising ways, as the quote from an expert in Project 3 (Pet care) explains:

> A cultural shift is happening in China. And it's also connected to other things that are happening, like

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

> the whole digital revolution and how (young) people are spending a lot of their lives on phones and on devices, and they're not going out as often. They're not having contact with friends or they're not going to restaurants or bars and doing those sorts of things. So, they're living a different life. They're living a life which is less connected with real people. So, when you think about people's relationships with pets, you have to understand all these other things that are changing at the same time.

On the other hand, our data show that the experts relied on explanatory abduction not only to infer plausible assumptions about specific unexpected user behaviors or technological changes, but also about more general unexpected phenomena related to the context of reference as a whole. For example, the attempt to provide a novel or alternative definition of a consolidated and well-known concept that no longer fits the current scenario, as this quote from Project 4 (Professional photography) illustrates:

> We are not saying that creativity is losing power, but that it is changing the way we use creativity on the set.

In addition, experts infer assumptions about how things might go as a result of a new scenario and try to expand the level of awareness, as this quote from Project 4 (Professional photography) shows:

> What I'll do if I'm photographing you is we'll shoot a couple of pictures, make sure the lighting looks good, and make sure everything's right. Make sure I didn't set a setting wrong or that, you know, your sweater looks good, all that. Once that's there, I'll talk to you while we're photographing, and I'll shoot through the whole thing. So, I'll ask you questions, you'll respond to me throughout that process. I'm taking pictures and that's partly why I'm taking more. Some of those pictures are going to be terrible because I'm going to hit the shutter right when you opened your mouth. But I could also get you right when you're thinking about your response to something or as you're laughing at something. And if I wait until I see that as well, I can miss that. And so being able to shoot through and interact with you as I'm taking pictures creates so many more opportunities for a really wonderful picture.

In all the situations described, the reasoning logic adopted responds to *what it might be*, as no frame of reference is available to the experts, who are forced to imagine and start reframing the problem to make sense of it (Maitlis and Christianson 2014). Therefore, explanatory abductive reasoning allows experts to move beyond the pure assessment activity, leading to dynamics that enable introducing new assumptions into the conversation (Sukhov et al. 2021). Table 7 provides selected quotes (see Appendix S1 for additional quotes).

4.5 | Speculative Innovative Abductive Reasoning (What It Might Become)

In addition to introducing new assumptions by inferring known rules that are new to the client's context of use and contrary to the hypothetical frame identified by the firm, our data show that experts even inferred entirely new rules to explain unattended facts by imagining unconventional new assumptions. Through innovative abduction, the external experts attempted to explain an unexpected observation by creatively introducing a new idea and then the assumption that, if true, supported the idea itself (Garbuio and Lin 2021). Therefore, in this future-oriented thinking, experts tried to imagine *what it might become* (see Table 8 and Appendix S1 for selected quotes). Our data show that external experts relied on innovative abductive reasoning in a speculative way, proactively imagining how the scenario might change and creating new images by recognizing contrasting signals or imagining possible futures.

In speculating about ideas and assumptions to explain unexpected observations, the experts seemed to follow different paths. On the one hand, they unleashed their imagination toward unconventional assumptions by imagining possible futures, as in these quotes from Project 4 (Professional photography):

> Probably, in the future it will be even stronger because nowadays we are looking for photographers that we can say are multitasking.

Or externalizing unorthodox thoughts:

> When you say there is good quality, what do you mean by good quality of a photo? You know it's hard to explain. Is the quality of the photo today the same that you would judge 10 years ago? Or is it something new?

Or externalizing new perspectives of the ongoing discussion, shifting the point of view, as in Project 2 (Haircare):

> So, they're figuring out who am I in this world, rather than who should I be to fit into this world?

On the other hand, speculative abductive reasoning also occurs by provoking debate with paradoxical assumptions through recognizing contrasting cues and proposing assumptions to resolve the paradox. Our data show that paradoxical assumptions emerged in two main ways. For example, introducing counter-intuitive thoughts about how things might become, as in a quote from Project 1 (Skincare):

> One day I might say "Hey, you know, I want to be a rock star," and then the next day I'm a rock star, but I also practice gardening. And that's fine. So, I feel that there's this big difference of having to be accepted versus making yourself accepted.

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 7 | Explanatory abductive reasoning, selected quotes.

Aggregate dimension: explanatory abductive reasoning	
Second-order themes	**Selected evidence on first-order categories**
Envisioning plausible assumptions that recognize unexpected user behaviors and technological changes	*Giving explanations for unexpected user behavior* Now they don't buy anything. They don't buy a house because it's difficult to buy a house. They don't buy a car because they rent it. They don't buy a computer because they rent it, and so on and so forth. So, I think there is more focus on building my life on myself, on my body, on my soul, on my capabilities. *Giving explanations for how technological changes influence user behavior* We put tools into artists' hands, and they go and do things that nobody would ever expect them to do. You can't anticipate what it is that somebody with a creative mindset is going to do in order to fulfill their vision or create something. Maybe it happens serendipitously. Maybe it happens with intent, maybe it happens over time. It's very much a creative process. *Giving explanations for what technological change can enable* Technology is the tool that can frame their schedule and of course frame their time from non-core activities, and is the great technological evolution of today.
Shaping plausible assumptions that recognize unexpected relations among phenomena	*Providing novel or alternative definitions for consolidated concepts* We believe that beauty somehow could be objective nowadays on the web because we can define what the right pixel is and we can draw the attention of customers even better. We are trying to, not automatize, but standardize the brief. *Trying to explain how things might go, expanding the level of awareness* I think the accessory is really useful to help with the flow right now. I think it's helpful with the flow because it allows you to have the camera in a safe position pointed in the right direction, in a way that I don't have to look through the camera, I don't have to hold it and have that physical barrier between me and my subject, which can be intimidating for people.

Paradoxical assumptions also emerged from identifying conceptual tensions, as in the quote from Project 4 (Professional photography):

> Pictures speak a thousand words, but now with photo manipulation to such a degree that you can change anything within the pixels of a photograph, it starts hitting on elements of truth.

4.6 | Moral Innovative Abductive Reasoning (What It Should Be)

In addition to speculative innovative abduction, external experts relied on abductive reasoning by consciously reconnecting with their values and beliefs, using what Pendleton-Jullian and Brown (2016) refer to as moral imagination, imagining *what it should be* (see Table 9 and Appendix S1 for selected quotes). We also observed that external experts tend not to stray too far from their core values and beliefs, almost as if making an ethical assessment of the situation. This did not prevent them from interpreting the novel context in new ways but led them to introduce a kind of moral reasoning, reflecting on how things should become and generating assumptions for a desirable future scenario. These assumptions emerged in two main ways. First, experts tended to bring the reflection back to their core values. In doing so, they questioned how the meaning of the established value is evolving, as in this quote from Project 2 (Haircare):

> The big question in fashion is "what is beauty?" and "what is authenticity?"

They also reflected on what is considered ethical in a continuous evolving landscape, as in this quote from Project 1 (Skincare):

> I think if a few years ago it was just wanting more visibility and making sure we had options for skin tone, just options to choose different ethnicity or different race for your character. Now there's really a push to make sure that we're being as equitable as we can and how we're portraying those characters, what the quality is and how they relate to other details about your character that you can customize. So, it's kind of a commitment that we make to players to continue to revisit it, and continue to bring them into the conversation.

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 8 | Speculative innovative abductive reasoning, selected quotes.

Aggregate dimension: speculative innovative abductive reasoning	
Second-order themes	**Selected evidence of first-order categories**
Igniting imagination with unconventional assumptions	*Imagining the future*
	What is it that people say about photography education in the future?
	Externalizing unorthodox thoughts
	And I found that because I was trying to find ways to intentionally corrupt a digital file. Sometimes with digital files, weird things will happen because something gets corrupted in the file. I was trying to figure out if there was a way to make that happen on purpose.
	Externalizing new perspectives
	Let's say we're photographing in your home. I haven't been there, so I don't know what it's like. And I have found that if you come in with too much of an idea of what you want, often you can't accomplish that in whatever space or with whatever person. And rather than allowing the creativity to flow and come up with a great idea because you're trying to force something to happen that doesn't work, you waste time instead of walking in and saying like "that window behind you is beautiful." I bet there's a cool shot by that window, but if I had this idea that I wanted to photograph you in your kitchen, maybe your kitchen looks terrible. I don't know. You're in Italy probably everybody's kitchen is great, but, um, so it's better if I walk in with more of an open mind in my experience.
Provoking debate with paradoxical assumptions	*Introducing counterintuitive thinking*
	We talk a lot about how we humanize pets but no about how they humanize us.
	Identifying emerging conceptual tensions
	When I am shooting, I am in the flow, but also when I am planning.

Second, they identified new pathways that align with the changing reality in a desirable way. This type of reasoning involved reflecting on how established pathways might evolve, as an expert from Project 4 (Professional photography) noted:

> It gets us thinking about where we might take things or what things might mean. So, what I did was to think about the changes that I've seen or the changes that our students are encountering so that I can think about how directions might change.

Alternatively, external experts reflected on what needs to be learned to pursue desirable pathways, as illustrated in the following quote from Project 3 (Pet care):

> Not everything fits the same because every pet is different. Some of the pets are happy to be alone, which is up to the owner because the owner will understand that. But some pets are not. So, innovation is needed to look after that, those breeds of pets where the owner doesn't actually know anything about the breed, but if you leave them alone for more than three hours, or if they don't hear your voice for three hours, they will go into emotional distress and then obviously don't eat, they just lie around. They just let go. So that, that is important as well to understand that there are different breeds of pets, because not everything fits everyone.

5 | Discussion

This study examines how external experts apply reasoning logics to assist firms reframe innovation problems. In the projects we observed, the top management teams' problem framing led to the development of preliminary hypothetical frames, albeit based on incomplete and inconsistent knowledge (Peirce 1998). The involvement of external experts allowed further reframing toward more reliable and plausible renewed frames (e.g., Garbuio and Lin 2021). The external experts were not tasked with assessing the hypothetical frames themselves, but their underlying assumptions. By starting with a given set of assumptions to evaluate, the experts provided not only a deeper understanding of the assumptions as input—some validated, others discarded—but also new assumptions that then led to the creation of renewed frames. In particular, through the external experts, the firms benefited from enriched and higher-order frames in terms of relevance and plausibility, which could then be easily translated into design principles, namely insights into the context and solid guidelines for developing the renewed frames into concrete solutions (Artusi et al. 2024).

By studying the reasoning logics emerging from the dialogues of the external experts involved in problem reframing, we uncovered how these logics are used and lead to the development of renewed frames by the firm. Our findings show that experts adopt both deductive and inductive logics, evaluating the observed (assumptions) in relation to the known, as well as abductive logics, generating new assumptions to explain the unexpected observations. We propose that problem reframing is an ongoing

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

process that involves the interplay of elaboration and generation dynamics. To illustrate, we developed the process model shown in Figure 2 and explained next.

5.1 | Elaborating Assumptions Through Reverse Deduction and Induction: Defining What We Know

Clearly emerging from our data is that the inverse deduction and induction reasoning logics are critical for problem reframing. This is curious and may sound counterintuitive, as problem reframing is conceived as an idea generation and opportunity identification activity (Cramer-Petersen et al. 2019; Dong et al. 2016; Eisenmann et al. 2012; Garbuio and Lin 2021). Consistently, scholars argue that abductive reasoning predominates in problem reframing as innovators move in the realm of the unknown, while deductive and inductive reasoning are more appropriate in the later stages of innovation when analytical and technical decisions are based on what is known (Cramer-Petersen et al. 2019; Garbuio and Lin 2021).

However, we observed that the deductive and inductive reasoning logics are crucial for activating the conversation, but in reverse, as the experts started from the assumptions themselves when evaluating them and connected them to known rules. Through reverse deductive and inductive reasoning, experts incorporate the assumptions from the hypothetical frame into their own interpretation schema, making them more familiar and understandable. When presented with a set of novel and unfamiliar assumptions, they shape and bring them into their own world to make them more accessible and manageable.

For example, in Project 2 an expert was presented with the assumption that today's Gen Zers seek support to try out and experiment with different situations in order to shape their identity. Reflecting on this, he began referencing a well-established concept from his field of expertise—education—stating "education now is a multidimensional experience, pushing boundaries." Right after he started to reframe the assumption by sharing: "I am eight years in my company now, and what I have definitely noticed is the lack of barriers that kids face - like going away for a year. The idea of going to Nebraska for a year to a local high school is challenging. It's rewarding, but they are not afraid, they see the value, they see the cultural immersion. I suppose it will redefine their identity or maybe it will be the definition of their identity because they know they will come back a different person." In this process, a preliminary reframing of the assumptions occurs. They started largely with reverse deduction, relying on theories proven in their own domain to reinforce the assumptions of the hypothetical frame, and then with reverse induction, relying on patterns emerging in their own domain to give concreteness to their understanding and interpretation of the underlying assumptions and to reinforce the hypothetical frame. In a sense, through reverse deduction and induction, the experts assessed the reliability and plausibility of the hypothetical frame by reframing it (Garbuio and Lin 2021). Specifically, reverse deduction refers to *what it must be*, while reverse induction refers to *what it is* (Roozenburg 1993; Dorst 2011). The former anchors reasoning to well-known and undeniable rules that anyone, even with different backgrounds, can validate and recognize as true, while the latter anchors reasoning to the realm of probability.

Thus, through inductive reasoning, experts rely on rules that may be true and meaningful to themselves but may also be false

TABLE 9 | Moral innovative abductive reasoning, selected quotes.

Aggregate dimension: moral innovative abductive reasoning	
Second-order themes	**Selected evidence of first-order categories**
Sparking reflections about founding values	*Questioning how the meaning of values is evolving*
	For me authenticity is about (...).
	Reflecting on what is considered as ethical today
	To think about the best way that photographers and photography can work in the world is to use those human attributes. So, to look at the ethical and the physical in terms of how we look at photography and where that might go we talk about digital, physical, and biological systems
Proposing desirable new pathways	*Reflecting on how established pathways might evolve*
	So, photography can be about truth, and it can be about fiction, and it can be about news, and it can be about entertainment, and it can be about business or science or art. It can be a document, it can be propaganda, it can be all of those things. So, you know, everybody uses that shorthand, you know, a picture tells a thousand words, but a picture also has a thousand meanings. And we just have to try and guide the meanings that we have through photography.
	Pondering what needs to be learned for desirable pathways
	I don't feel very comfortable when you treat a pet like a human, i.e., I'm a bit against dressing them up, putting them in clothes. So I think a pet, you should understand what they are naturally in their natural environment. You know, let a pet be a pet, let a human be a human, we are both animals, and we can be part of the same family. We can live together in an amazing way.

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

or not fully acceptable to others. In this way, external experts pave the way for novel insights: despite reframing the assumptions to their own domain, the diversity among external experts (and relative domains) leads to the emergence of different interpretations of these assumptions, questioning their validity and introducing a degree of uncertainty into the conversation that needs to be explained, leading to further reframing (Bianchi and Verganti 2021).

5.2 | Generating Novel Assumptions Through Explanatory and Innovative Abduction: Paving the Way to the Renewed Frame

Although the introduction of new elements may create uncertainty, it ultimately leads to a richer and more meaningful understanding (Bianchi and Verganti 2021; Stigliani and Ravasi 2012; Schön 1983). In the projects we analyzed, experts progressively engaged in bringing new elements into the conversation. Diversity among experts allows different frames to be compared, opening up new possibilities (Ananth et al. 2024). By broadening their perspectives through listening to others, experts draw on alternative reasoning logics, making new associations and assumptions (Cramer-Petersen et al. 2019; Stumpf and McDonnell 2002).

The new assumptions are first approached through explanatory abduction, reflecting on theories and patterns from the expert's own domain that contrast with the assumptions of the hypothetical frame, and then reasoning about what is true and plausibly adaptable in the new problem space (*what it might be*). For instance, continuing the example in Project 2, he furthered his reflection by stating, "There is a big drop in their (Gen Zers') happiness, that emerges when they return (from an experience abroad). They obviously miss the place they were in, they missed the life they had. They probably miss much of the person they were able to be. And then when they came back to their usual Monday-to-Friday routine at their local school, they likely feel as though the adventure has been lost. I think that this feeling starts to open up their world to: 'ok, where is the next adventure?'". In this way he begins to move beyond the idea that having different experience merely shapes personal identity and instead connects it to the themes of adventure and personal exploration—an insight that emerged from the reasoning of another expert in the same panel. Explanatory abduction and reverse induction are closely related: the second-order themes of both aggregate dimensions refer to similar elements (e.g., user behavior, specific instances), but use different reasoning logics. The process of opening up to new assumptions is gradual: it starts incrementally from what has been drafted through inductive reasoning (*what it is*), and then evolves radically up to innovative abduction (*what it might become*).

Opening up to new assumptions leads experts to begin to speculatively dare by imagining unconventional or paradoxical reasoning. In this way, they bring unexpected and breakthrough possibilities about a possible future, or even introducing unorthodox or counterintuitive reflections that push the boundaries of imagination beyond what is possible or socially acceptable. The expert in the previous example did exactly this, extending his reasoning even beyond what he had heard from the other expert—shifting from personal identity to adventure and personal exploration—toward the idea of personal transformation by stating: "The exploration side, I think, is emerging when getting out of your comfort zone and start that inner exploration: 'who am I going to be when I get into a different situation? What is it going to reveal about myself?'".

At the same time, speculative innovative abduction involves the recognition of responsibility, whereby imagination is no longer merely speculative, but evolves toward a moral dimension (Pendleton-Jullian and Brown 2016). Thus, experts tap into their will to interpret and imagine desirable new assumptions. Through such reasoning, the emerging assumptions fall into the realm of *what it should become*. In relation to this, the expert in Project 2 expresses the will to refer to this transformation process as a process of "becoming," which is not just about keep experiencing and transforming as much as possible, but it is about engaging in a process of self-awareness and consciousness.

Once new insights emerge, the process can begin anew, with these insights becoming a fresh set of elements to explore. In our data, each expert engaged in the infinite loop represented in Figure 2, building not only on the assumptions received as input but also on the insights that surfaced through other experts' reasoning. In this sense, each loop starts from an initial assumption yet becomes progressively shaped by the reasoning developed in preceding loops. In this way, reframing the innovation problem can lead to continuous and further refinements of the problem space. Experts contribute to this process as external helpers (Fisher et al. 2018), dedicating their time and effort to assist firms in reframing the innovation problem. By providing new insights, experts help firms create new assumptions and a renewed frame that goes beyond the initial perspective (Ainamo et al. 2021).

Our findings show that this is an interpretation process in which experts and firms collaborate and reason together. Engaging external experts is not a one-time event, but an intensive process of sensemaking that aims not only to clarify what the innovation problem is (problem framing) (Ananth et al. 2024), but more importantly, to generate a new perspective and make new connections through problem reframing. Reasoning logics play a key role in highlighting the mechanisms that facilitate problem reframing and provide plausible explanations for the nuanced social processes involved (Reynolds et al. 2023).

5.3 | Theoretical Implications

Our paper contributes to two literature streams: problem reframing and reasoning logics.

As mentioned, several studies have investigated alternative perspectives of problem reframing (Reynolds et al. 2023; Bianchi and Verganti 2021; Stigliani and Ravasi 2012; Rosenbloom et al. 2016). Our study shows that external experts can contribute significantly to problem reframing not only by providing new knowledge, as extensively demonstrated (Chesbrough 2003, 2006, 2010; Chesbrough and Bogers 2014), but also by helping make sense of existing knowledge, leading to a new and more

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

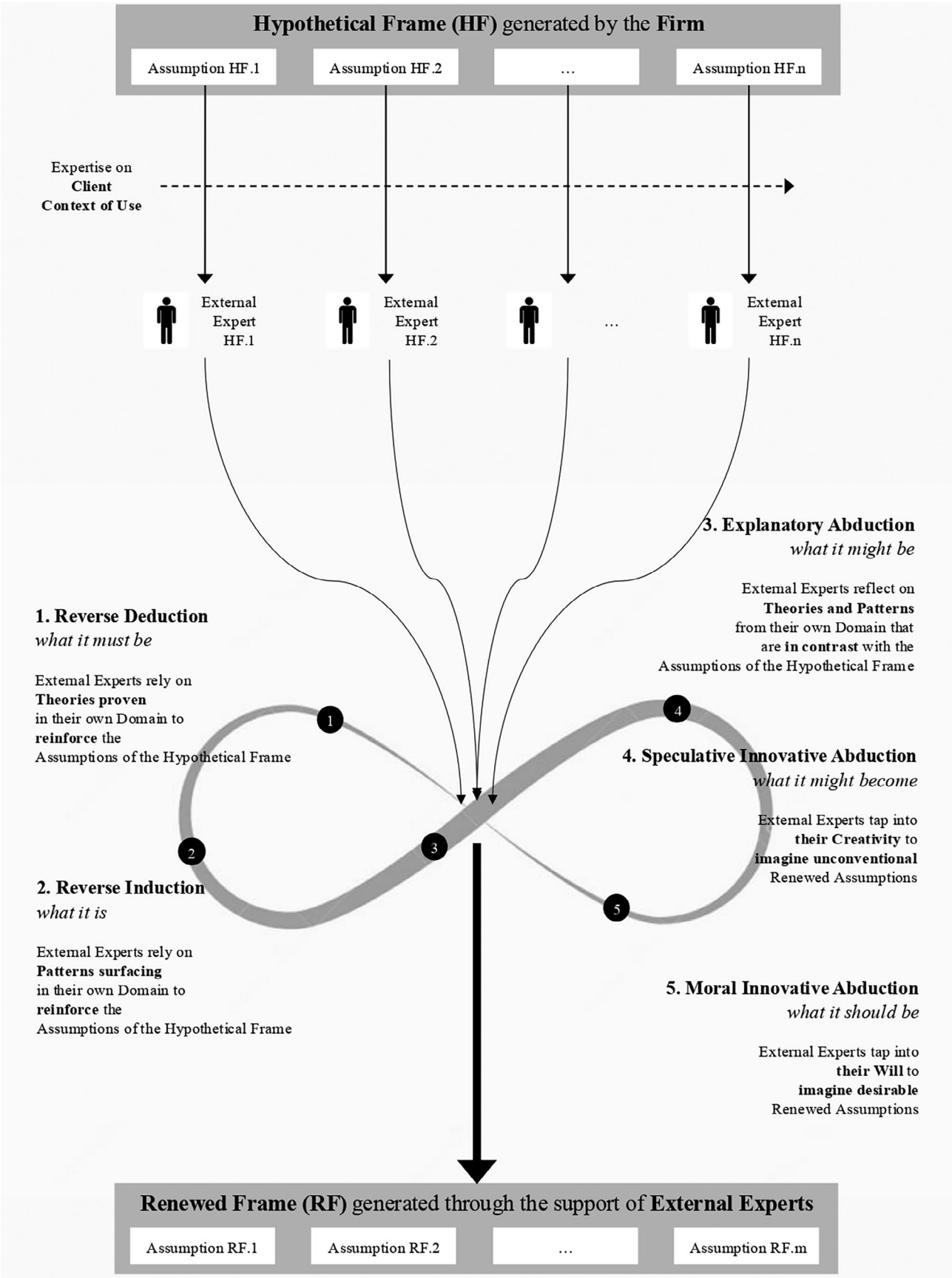

FIGURE 2 | The reasoning logics used by external experts in reframing innovation problems.

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

meaningful problem. To this end, the adoption of reasoning logics allows for a proper exploration of the problem reframing. Starting from a given set of assumptions to be evaluated, the reasoning logics adopted by experts allow reframing the assumptions as input, and then generating new assumptions based on new connections between the frames and concepts, leading to a reframing of the entire problem space.

Furthermore, scholars argue that research often downplays the cognitive and emotional mechanisms through which frames are socially constructed and negotiated (Ananth et al. 2024; Bellis, Magnanini, et al. 2024; Kaplan 2008; Cornelissen and Werner 2014; Fisher et al. 2018). Cornelissen and Werner (2014) suggest that empirical studies should more closely examine reframing as an ongoing process of meaning construction. Ananth et al. (2024) and Reynolds et al. (2023) also call for a deeper exploration of the mechanisms and key activities involved in problem reframing. Our study addresses these calls by focusing on problem reframing activities through the theoretical lens of reasoning logics. Specifically, we show how external experts play a critical role in helping firms overcome the rigidities of their existing frames and facilitate the development of renewed frames, as well as the mechanisms that enable this reframing process.

From a reasoning logic perspective, our study shows that deductive and inductive reasoning are not only particularly useful in the late stages of innovation when analytical and technical decisions are made based on what is known (Cramer-Petersen et al. 2019; Garbuio and Lin 2021), but can also be critical to fostering early conversation when practiced appropriately. We observed that experts engage in the reframing process through reverse deductive and inductive reasoning, bringing the assumptions into their own interpretive schema to make them more familiar and understandable, thus initiating the reframing of the problem. In particular, each expert enters the conversation primed with his or her own perspective. Reverse deduction and induction are attempts to create a common ground among all the experts, where their diversity stimulates the search for new interpretations through abduction. Explanatory abductive reasoning plays a central role by allowing individuals to project their own perspectives onto a different context, thereby generating new insights and elevating the reframing activity to a deeper level of reflection. While previous studies emphasize the role of innovative abduction in imaging new hypotheses (Chiffi and Pietarinen 2020; Cunningham 1998), our study highlights the nuances of two different types of innovative abduction. Speculative, innovative abduction focuses on imagining unconventional or paradoxical assumptions that open up unexpected, breakthrough possibilities about a possible future. In contrast, moral innovative abduction relies on values that experts perceive as good and meaningful for the world.

The discovery of these two types of abduction has important implications for innovation studies. While research often emphasizes breakthrough and radical innovation (Salter et al. 2014), our findings highlight an important nuance: for an assumption to be plausible and relevant, it also undergoes a moral evaluation. This shows that innovation is not only about making bold, speculative leaps, but also about aligning new ideas with ethical and societal values. This offers a richer perspective on what makes innovation both impactful and sustainable in practice.

Finally, the framework in Figure 2 illustrates the interplay between the different reasoning logics in problem reframing: reverse deduction, reverse induction, explanatory abduction, speculative and moral innovative abduction. While several studies have explored the role of individual reasoning logics in addressing innovation problems (Dong et al. 2016; Eisenmann et al. 2012; Garbuio and Lin 2021), few have examined how reasoning logics emerge and the social mechanisms they enable, calling for further research (Ananth et al. 2024; Reynolds et al. 2023; Fisher et al. 2018). Our study responds to this call by highlighting the synergies, complementarities, and contributions of different reasoning logics in the process of problem reframing.

5.4 | Managerial Implications

Our study also offers valuable insights for managers. First, it demonstrates that the value of involving external experts lies not only in their knowledge or domain expertise, but also in the ways they interact and reason throughout the innovation process. What emerges is a dynamic reframing process in which experts progressively build on, challenge, and reconfigure each other's assumptions. These subtle reasoning moves, often invisible to managers and perceived as nonlinear and unpredictable, shape the evolution of the problem more than any single idea or insight. For managers, this means that attention should not focus solely on the breakthrough solutions proposed by external experts. Instead, managers need to listen closely to *how* experts reason, how their arguments shift over time, and how these shifts trigger new reframing trajectories. Actively engaging in this evolving reasoning process is essential for the emergence of more plausible and robust frames.

While managers are themselves experts, their perspective may be constrained by the organizational routines and existing narratives. External experts offer complementary views that help broaden and reframe the scope of the problem and make alternative frames plausible. However, it is only through shared engagement in the reframing process that a more complete picture can emerge and a renewed frame can be identified. It is through this joint engagement, rather than through isolated expert inputs, that problem reframing can ultimately emerge.

Second, our study underscores the importance of creating spaces that allow external experts to engage in iterative reflection and build on one another's reasoning, both of which are fundamental to enabling problem reframing. In the context of our research, this space was facilitated through a series of carefully curated sessions organized by Alpha. These sessions were intentionally designed to encourage deep thinking, allow for perspective sharing, and support the gradual unfolding of reasoning patterns over time. Although the sessions in our study were conducted virtually, we argue that in-person interactions—especially those situated in neutral or informal environments—could offer even greater potential for fostering cognitive flexibility and psychological safety. Managers could actively consider organizing off-site retreats, immersive workshops, or other forms of dialogic encounters in spaces removed from day-to-day operations. The goal is not to stage one-off brainstorming events, but rather to establish reflective settings that promote

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

collective sensemaking and recursive reasoning. Such spaces can become catalysts for reframing, particularly when they are structured to enable alternative viewpoints, challenge dominant assumptions, and support the emergence of new frames that are not immediately constrained by organizational routines or pre-existing logics. Importantly, these spaces should be protected from the pressure to deliver immediate solutions; instead, they should cultivate a longer-term mindset focused on exploration, iteration, and mutual learning.

Third, the findings of our study point to the critical role of managerial openness to diverse and at times uncomfortable forms of feedback. Reframing requires more than just listening to external experts; it involves a willingness to reconsider, reconfigure, and even let go of existing assumptions and frames. External experts, by virtue of their distance from internal logics and routines, often bring in reasoning patterns that challenge established mental models. Managers should move beyond simply seeking validation or incremental improvement of their initial hypotheses. Instead, they should position themselves as co-designers of meaning, actively engaging with the tensions, disagreements, and novel insights that external experts introduce. The reframing process is not about finding the "correct" frame but about surfacing what has previously been overlooked, ignored, or deemed too unconventional. As such, reframing becomes a process of deepening understanding with each iteration—one that invites managers to embrace uncertainty not as a threat but as a resource for innovation. In this sense, feedback is not an endpoint but a generative force that, when engaged with openly and iteratively, can lead to the co-construction of renewed and meaningful directions for innovation.

Finally, our study highlights the transformative power of dialogue as a tool for co-creation and collective learning. Dialogue is not merely an exchange of opinions or information; it is a generative process through which actors can engage in inquiry, exercise critical thinking, and transcend the limitations of their existing frames. Dialogue—anchored in curiosity, openness, and mutual respect—creates the conditions for cognitive flexibility and conceptual expansion. It allows individuals to suspend judgment, entertain alternative interpretations, and reflect more deeply on the assumptions that underlie their decisions and strategies. As such, it becomes a powerful vehicle for both personal and organizational transformation. In line with the opening quote of this manuscript, we invite managers not only to initiate dialogue with external experts but also to fully immerse themselves in it. True dialogue requires the willingness to be changed by what one hears. It is through this kind of engagement that managers can begin to imagine alternative frames of the world and formulate hypotheses about where the world might be heading.

5.5 | Limitations and Future Research

Like any study, ours has certain limitations that also offer promising avenues for future research. First of all, our study focuses on the reasoning logics employed during the problem reframing process, without assessing the quality or impact of the resulting assumptions and renewed frames. While we trace how firms move from hypothetical to renewed frames, we do not evaluate the goodness of these new frames in terms of their novelty, coherence, or meaningfulness. Future research could explore how different reasoning logics affect the quality of the reframed problem, investigating, for example, whether certain reasoning patterns are more likely to generate breakthrough, actionable, or strategically valuable insights.

In order to examine the contribution of external experts during reframing processes, our analysis focused on the transformation from a hypothetical to a renewed frame. However, our cases did not include external experts engaged at the very inception of the innovation process. Future studies could investigate alternative trajectories—particularly in scenarios where external experts are involved earlier, even during the initial formation of the hypothetical frame. This may uncover different pathways and dynamics in the evolution of problem reframing.

As shown in Table A1 (Appendix S1), the expert groups involved in the various projects were heterogeneous. While this heterogeneity enriches our findings, it also raises important questions about composition. Future research could examine which combinations or configurations of external expertise are most conducive to meaningful problem reframing. For example, under what conditions do cognitive diversity, industry background, or experiential distance contribute most effectively to innovation problem reframing.

We adopted an ethnographic approach to observe the reasoning logics used by external experts, without attempting to influence or steer their application. While this approach provided an authentic account of how these logics unfold, it did not allow us to examine how specific practices or interventions might stimulate or support particular reasoning patterns. Future studies could explore the mechanisms—facilitation techniques, toolkits, team dynamics, or contextual cues—that encourage or inhibit the use of specific reasoning logics, potentially enhancing their practical utility.

Although our data are rich and drawn from multiple projects, we relied on a single methodological framework developed and implemented by the consulting firm. This consistency allowed us to ensure comparability and depth of insight, particularly regarding the reasoning logics, but it may also limit the generalizability of our findings. Future research could broaden the scope by examining problem reframing across a wider variety of organizational contexts, industries, and methodological approaches.

All the interactions with external experts in our study took place in a virtual environment. Existing literature suggests that digital media may constrain interpersonal dynamics, reducing spontaneity, embodied cognition, and intuitive responses that are often important in early-stage innovation (Bellis et al. 2022, Bellis, Cunial, et al. 2024). Future research should investigate the reasoning logics of external experts in physical, co-located settings to assess how the medium of interaction shapes the reframing process and the nature of the contributions.

Our study focused primarily on the cognitive dimension of reframing—specifically on the reasoning logics employed by

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

external experts. While this focus allowed us to unpack important cognitive mechanisms, future studies could explore other dimensions, such as motivational, emotional, or social mechanisms that may also influence how innovation problems are reframed (Ananth et al. 2024). These additional layers could offer a more holistic understanding of how external contributions shape the reframing process.

In conclusion, while our study provides new insights into how external experts reason when assisting firms in reframing innovation problems, it also opens up several directions for future research. Continued exploration of these themes can deepen our understanding of problem reframing and ultimately support firms in navigating innovation challenges in more adaptive, reflective, and transformative ways.

Acknowledgments

The authors would like to express their sincere gratitude to the anonymous reviewers and the editors for their dedication, constructive feedback, and continuous support throughout the review process. Their insightful comments and valuable suggestions have been crucial in significantly improving the quality, clarity, and rigor of this manuscript. We deeply appreciate the time and expertise they devoted to strengthening our work. Open access publishing facilitated by Politecnico di Milano, as part of the Wiley - CRUI-CARE agreement.

Funding

The authors have nothing to report.

Ethics Statement

The authors have read and agreed to the Committee on Publication Ethics (COPE) international standards for authors.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data available on request due to privacy/ethical restrictions. A portion of the data that supports the findings of this study is available in the Supporting Information of this article.

Endnotes

[1] https://www.youtube.com/watch?v=hAiIskxPz1Y&ab_channel=Slush.

References

Abdulla, A. M., S. H. Paek, B. Cramond, and M. A. Runco. 2020. "Problem Finding and Creativity: A Meta-Analytic Review." *Psychology of Aesthetics, Creativity, and the Arts* 14, no. 3: 3–14. https://doi.org/10.1037/aca0000194.

Ainamo, A., C. Dell'Era, and R. Verganti. 2021. "Radical Circles and Visionary Innovation: Angry Birds and the Transformation of Video Games." *Creativity and Innovation Management* 30, no. 3: 439–454. https://doi.org/10.1111/caim.12458.

Altuna, N., C. Dell'Era, P. Landoni, and R. Verganti. 2017. "Developing Radically New Meanings Through the Collaboration With Radical Circles: Slow Food as a Platform for Envisioning Innovative Meanings." *European Journal of Innovation Management* 20, no. 2: 269–290. https://doi.org/10.1108/EJIM-06-2015-0045.

Amabile, M. T. 1988. "A Model of Creativity and Innovation in Organizations." *Research in Organizational Behavior* 10, no. 1: 123–167.

Ananth, P., M. Baer, and D. Deichmann. 2024. "Developing Problem Representations in Organizations: A Synthesis Across Literatures and an Integrative Framework." *Journal of Management* 51, no. 6: 2212–2244. https://doi.org/10.1177/01492063241291532.

Artusi, F., P. Bellis, and R. Verganti. 2024. "When Products Speak Differently: Designing New Languages for Established Products." *Creativity and Innovation Management* 34, no. 2: 297–311. https://doi.org/10.1111/caim.12637.

Baer, M., K. T. Dirks, and J. A. Nickerson. 2013. "Microfoundations of Strategic Problem Formulation." *Strategic Management Journal* 34, no. 2: 197–214. https://doi.org/10.1002/smj.2004.

Barczak, G., A. Griffin, and K. B. Kahn. 2009. "Perspective: Trends and Drivers of Success in NPD Practices: Results of the 2003 PDMA Best Practices Study." *Journal of Product Innovation Management* 26, no. 1: 3–23. https://doi.org/10.1111/j.1540-5885.2009.00331.x.

Beckman, S. L. 2020. "To Frame or Reframe: Where Might Design Thinking Research Go Next?" *California Management Review* 62, no. 2: 144–162. https://doi.org/10.1177/0008125620906620.

Beckman, S., and M. Barry. 2009. "Design and Innovation Through Storytelling." *International Journal of Innovation Science* 1, no. 4: 151–160. https://doi.org/10.1260/1757-2223.1.4.151.

Bellis, P., M. Cunial, and D. Trabucchi. 2024. "Mastering Hybrid Worlds Through Digital Leadership: The Role of Agility in Fostering Innovation." *Business Horizons* 67, no. 4: 369–380. https://doi.org/10.1016/j.bushor.2024.04.002.

Bellis, P., S. Magnanini, and R. Verganti. 2024. "Dialogue for Strategy Implementation: How Framing Processes Enable the Evolution of New Opportunities." *Journal of Knowledge Management* 28, no. 11: 1–32. https://doi.org/10.1108/JKM-01-2023-0064.

Bellis, P., D. Trabucchi, T. Buganza, and R. Verganti. 2022. "How Do Human Relationships Change in the Digital Environment After COVID-19 Pandemic? The Road Towards Agility." *European Journal of Innovation Management* 25, no. 6: 821–849. https://doi.org/10.1108/EJIM-02-2022-0093.

Berg, J. M. 2019. "When Silver Is Gold: Forecasting the Potential Creativity of Initial Ideas." *Organizational Behavior and Human Decision Processes* 154: 96–117. https://doi.org/10.1016/j.obhdp.2019.08.004.

Bianchi, M., and R. Verganti. 2021. "Entrepreneurs as Designers of Problems Worth Solving." *Journal of Business Venturing Design* 1, no. 1–2: 100006. https://doi.org/10.1016/j.jbvd.2022.100006.

Bouschery, S. G., V. Blazevic, D. Roberts, and F. T. Piller. 2022. "Artificial Intelligence as an Actor in Innovation Teams: An Assessment of the GPT-3 Language Model." *Academy of Management Proceedings* 1: 11505. https://doi.org/10.5465/AMBPP.2022.11505.

Burke, K. 1937. *Attitudes Toward History*. New Republic.

Capaldo, A. 2007. "Network Structure and Innovation: The Leveraging of a Dual Network as a Distinctive Relational Capability." *Strategic Management Journal* 28, no. 6: 585–608. https://doi.org/10.1002/smj.621.

Chesbrough, H. 2003. *Open Innovation: The New Imperative for Creating and Profiting From Technology*. Harvard Business Press.

Chesbrough, H. 2006. *Open Business Models: How to Thrive in the New Innovation Landscape*. Harvard Business Press.

Chesbrough, H. 2010. "Business Model Innovation: Opportunities and Barriers." *Long Range Planning* 43, no. 2–3: 354–363. https://doi.org/10.1016/j.lrp.2009.07.010.

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Chesbrough, H., and M. Bogers. 2014. "Explicating Open Innovation: Clarifying an Emerging Paradigm for Understanding Innovation." In *New Frontiers in Open Innovation*, edited by H. Chesbrough, W. Vanhaverbeke, and J. West, 3–28. Oxford University Press.

Chiffi, D., and A.-V. Pietarinen. 2020. "Abductive Inference Within a Pragmatic Framework." *Synthese* 197: 2507–2523. https://doi.org/10.1007/s11229-018-1824-6.

Cornelissen, J. P., and M. D. Werner. 2014. "Putting Framing in Perspective: A Review of Framing and Frame Analysis Across the Management and Organizational Literature." *Academy of Management Annals* 8, no. 1: 181–235. https://doi.org/10.5465/19416520.2014.875669.

Cramer-Petersen, C. L., B. T. Christensen, and S. Ahmed-Kristensen. 2019. "Empirically Analysing Design Reasoning Patterns: Abductive-Deductive Reasoning Patterns Dominate Design Idea Generation." *Design Studies* 60: 39–70. https://doi.org/10.1016/j.destud.2018.10.001.

Cukier, K., V. Mayer-Schönberger, and F. de Véricourt. 2021. *Framers: Human Advantage in an Age of Technology and Turmoil*. Dutton.

Cunningham, D. J. 1998. "Cognition as Semiosis: The Role of Inference." *Theory & Psychology* 8, no. 6: 827–840. https://doi.org/10.1177/0959354398086006.

Dong, A., M. Garbuio, and D. Lovallo. 2016. "Generative Sensing: A Design Perspective on the Microfoundations of Sensing Capabilities." *California Management Review* 58, no. 4: 97–117. https://doi.org/10.1525/cmr.2016.58.4.97.

Dorst, K. 2011. "The Core of 'Design Thinking' and Its Application." *Design Studies* 32, no. 6: 521–532. https://doi.org/10.1016/j.destud.2011.07.006.

Dorst, K. 2015. *Frame Innovation: Create New Thinking by Design*. MIT Press.

Dunne, D., and D. Dougherty. 2016. "Abductive Reasoning: How Innovators Navigate in the Labyrinth of Complex Product Innovation." *Organization Studies* 37, no. 2: 131–159.

Eisenmann, T., E. Ries, and S. Dillard. 2012. *Hypothesis-Driven Entrepreneurship: The Lean Startup*. Harvard Business School.

Eling, K., and C. Herstatt. 2017. "Managing the Front End of Innovation—Less Fuzzy, Yet Still Not Fully Understood." *Journal of Product Innovation Management* 34, no. 6: 864–874. https://doi.org/10.1111/jpim.12415.

Emerson, R. M., R. I. Fretz, and L. L. Shaw. 1995. *Writing Ethnographic Fieldnotes*. Chicago University Press.

Fairhurst, G. T. 2005. "Reframing the Art of Framing: Problems and Prospects for Leadership." *Leadership* 1, no. 2: 165–185. https://doi.org/10.1177/1742715005051857.

Fetterman, D. M. 1998. *Ethnography*. Sage Publications.

Fetterman, D. M. 2019. *Ethnography: Step-By-Step*. Sage Publications.

Fischer, H. R. 2001. "Abductive Reasoning as a Way of Worldmaking." *Foundations of Science* 6, no. 4: 361–383. https://doi.org/10.1023/A:1011671106610.

Fisher, C. M., J. Pillemer, and T. M. Amabile. 2018. "Deep Help in Complex Project Work: Guiding and Path-Clearing Across Difficult Terrain." *Academy of Management Journal* 61, no. 4: 1524–1553. https://doi.org/10.5465/amj.2016.0207.

Fjeldstad, Ø. D., C. C. Snow, R. E. Miles, and C. Lettl. 2012. "The Architecture of Collaboration." *Strategic Management Journal* 33, no. 6: 734–750. https://doi.org/10.1002/smj.1968.

Folger, R., and C. Stein. 2017. "Abduction 101: Reasoning Processes to Aid Discovery." *Human Resource Management Review* 27, no. 2: 306–315. https://doi.org/10.1016/j.hrmr.2016.08.007.

Garbuio, M., and N. Lin. 2021. "Innovative Idea Generation in Problem Finding: Abductive Reasoning, Cognitive Impediments, and the Promise of Artificial Intelligence." *Journal of Product Innovation Management* 38, no. 6: 701–725. https://doi.org/10.1111/jpim.12602.

Gemser, G., G. Calabretta, and I. Karpen. 2025. "Co-Creating the Future Through Design Thinking: Deconstructing the Consumer co-Creation Process." *Journal of Product Innovation Management* 42, no. 3: 528–556.

Gioia, D. A., K. G. Corley, and A. L. Hamilton. 2013. "Seeking Qualitative Rigor in Inductive Research: Notes on the Gioia Methodology." *Organizational Research Methods* 16, no. 1: 15–31. https://doi.org/10.1177/1094428112452151.

Giorgi, S. 2017. "The Mind and Heart of Resonance: The Role of Cognition and Emotions in Frame Effectiveness." *Journal of Management Studies* 54, no. 5: 711–738.

Gonzalez, M. E. Q., and P. Haselager. 2005. "Creativity: Surprise and Abductive Reasoning." *Semiotica* 153: 325–342. https://doi.org/10.1515/semi.2005.2005.153-1-4.325.

Gottschall, J. 2013. *The Storytelling Animal: How Stories Make Us Human*. Mariner.

Hammedi, W., A. C. R. van Riel, and Z. Sasovova. 2011. "Antecedents and Consequences of Reflexivity in New Product Idea Screening." *Journal of Product Innovation Management* 28, no. 5: 662–679. https://doi.org/10.1111/j.1540-5885.2011.00831.x.

Hymes, D. 1974. *Foundations of Sociolinguistics: An Ethnographic Approach*. University of Pennsylvania.

Isaacs, W. 1999. *Dialogue and the Art of Thinking Together*. Currency.

Jepsen, L. B., C. Dell'Era, and R. Verganti. 2014. "The Contributions of Interpreters to the Development of Radical Innovations of Meanings: The Role of 'Pioneering Projects' in the Sustainable Buildings Industry." *R&D Management* 44, no. 1: 1–17. https://doi.org/10.1111/radm.12035.

Johnson-Laird, P. N. 2008. "How We Reason: A View From Psychology." *Reasoner* 2: 4–5.

Kahneman, D., and A. Tversky. 1979. "On the Interpretation of Intuitive Probability: A Reply to Jonathan Cohen." *Cognition* 7, no. 4: 409–411. https://doi.org/10.1016/0010-0277(79)90024-6.

Kaplan, S. 2008. "Framing Contests: Strategy Making Under Uncertainty." *Organization Science* 19, no. 5: 729–752. https://doi.org/10.1287/orsc.1070.0340.

Kaplan, S., and K. Vakili. 2015. "The Double-Edged Sword of Recombination in Breakthrough Innovation." *Strategic Management Journal* 36, no. 10: 1435–1457. https://doi.org/10.1002/smj.2294.

Kim, W. C., and R. Mauborgne. 2004. "Value Innovation." *Harvard Business Review* 82, no. 7/8: 172–180.

LeCompte, M. D., and J. J. Schensul. 1999. *Designing and Conducting Ethnographic Research*. Rowman Altamira.

Locke, K. 2001. *Grounded Theory in Management Research*. Sage Publications.

Magistretti, S., C. T. A. Pham, and C. Dell'Era. 2025. "The Creative Process of Problem Framing for Innovation: An Integrative Review and Research Agenda." *Journal of Product Innovation Management* 42: 987–1018. https://doi.org/10.1111/jpim.12783.

Magnani, L. 1995. "Creative Processes in Scientific Discovery." *European Journal of High Ability* 6, no. 2: 160–169. https://doi.org/10.1080/0937445940060260.

Magnani, L. 2016. "The Eco-Cognitive Model of Abduction Ii: Irrelevance and Implausibility Exculpated." *Journal of Applied Logic* 15: 94–129. https://doi.org/10.1016/j.jal.2016.02.001.

Maitlis, S., and M. Christianson. 2014. "Sensemaking in Organizations: Taking Stock and Moving Forward." *Academy of Management Annals* 8, no. 1: 57–125. https://doi.org/10.5465/19416520.2014.873177.

March, L. 1976. *The Architecture of Form*. Cambridge University Press.

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Mueller, J. S., C. J. Wakslak, and V. Krishnan. 2014. "Construing Creativity: The How and Why of Recognizing Creative Ideas." *Journal of Experimental Social Psychology* 51: 81–87. https://doi.org/10.1016/j.jesp.2013.11.007.

Mumford, M. D., R. Reiter-Palmon, and M. R. Redmond. 1994. "Problem Construction and Cognition: Applying Problem Representations in Ill-Defined Domains." In *Problem Finding, Problem Solving, and Creativity*, edited by M. A. Runco, 3–39. Ablex Publishing.

Nonaka, I., and H. Takeuchi. 1995. *The Knowledge Creating Company*. Oxford University Press.

O'Connor, G. C., and A. D. Ayers. 2005. "Building a Radical Innovation Competency." *Research-Technology Management* 48, no. 1: 23–31. https://doi.org/10.1080/08956308.2005.11657292.

Patrucco, A. S., P. Bellis, D. Trabucchi, and T. Buganza. 2024. "Behavioral Biases and Cognitive Pitfalls: Navigating Resource Orchestration in Supplier-Partnered Innovation Projects." *IEEE Transactions on Engineering Management* 72: 227–239.

Paul, R., and L. Elder. 2013. *Critical Thinking: Tools for Taking Charge of Your Professional and Personal Life*. Pearson Education.

Peirce, C. S. 1998. *The Essential Peirce: Selected Philosophical Writings. 1893–1913. Volume 2*. Indiana University Press.

Pendleton-Jullian, A., and J. S. Brown. 2016. *Pragmatic Imagination*. Blurb.

Pham, C. T. A., S. Magistretti, and C. Dell'Era. 2023. "How Do You Frame Ill-Defined Problems? A Study on Creative Logics in Action." *Creativity and Innovation Management* 32, no. 3: 493–516. https://doi.org/10.1111/caim.12543.

Posen, H. E., T. Keil, S. Kim, and F. D. Meissner. 2018. "Renewing Research on Problemistic Search—A Review and Research Agenda." *Academy of Management Annals* 12, no. 1: 208–251. https://doi.org/10.5465/annals.2016.0018.

Reichertz, J. 2010. "Abduction: The Logic of Discovery of Grounded Theory." *Forum: Qualitative Social Research* 11, no. 1. https://doi.org/10.17169/fqs-11.1.1412.

Reiter-Palmon, R. 2017. "The Role of Problem Construction in Creative Production." *Journal of Creative Behavior* 51, no. 4: 323–326. https://doi.org/10.1002/jocb.202.

Reynolds, O., A. O'Dochartaigh, E. Secchi, D. Marshall, and A. Prothero. 2023. "Framing Innovation Success, Failure, and Transformation: A Systematic Literature Review." *Journal of Product Innovation Management* 42: 194–219. https://doi.org/10.1111/jpim.12706.

Richardson, R., and E. H. Kramer. 2006. "Abduction as the Type of Inference That Characterizes the Development of a Grounded Theory." *Qualitative Research* 6, no. 4: 497–513. https://doi.org/10.1177/1468794106068019.

Roozenburg, N. F. M. 1993. "On the Pattern of Reasoning in Innovative Design." *Design Studies* 14, no. 1: 4–18. https://doi.org/10.1016/S0142-694X(05)80002-X.

Rosenbloom, D., H. Berton, and J. Meadowcroft. 2016. "Framing the Sun: A Discursive Approach to Understanding Multi-Dimensional Interactions Within Socio-Technical Transitions Through the Case of Solar Electricity in Ontario, Canada." *Research Policy* 45, no. 6: 1275–1290. https://doi.org/10.1016/j.respol.2016.03.012.

Salter, A., P. Criscuolo, and A. L. J. Ter Wal. 2014. "Coping With Open Innovation: Responding to the Challenges of External Engagement in R&D." *California Management Review* 56, no. 2: 77–94. https://doi.org/10.1525/cmr.2014.56.2.77.

Schein, E. H. 1999. "Empowerment, Coercive Persuasion and Organizational Learning: Do They Connect?" *Learning Organization* 6, no. 4: 163–172. https://doi.org/10.1108/09696479910280622.

Scheufele, D. 1999. "Framing as a Theory of Media Effects." *Journal of Communication* 49, no. 1: 103–122. https://doi.org/10.1111/j.1460-2466.1999.tb02784.x.

Schön, D. A. 1983. *The Reflective Practitioner: How Professionals Think in Action*. Basic Books.

Schurz, G. 2008. "Patterns of Abduction." *Synthese* 164, no. 2: 201–234. https://doi.org/10.1007/s11229-007-9223-4.

Seidel, V. P. 2007. "Concept Shifting and the Radical Product Development Process." *Journal of Product Innovation Management* 24, no. 6: 522–533. https://doi.org/10.1111/j.1540-5885.2007.00269.x.

Senge, P. M. 1991. "The Fifth Discipline: The Art and Practice of the Learning Organization." *Performance and Instruction* 30, no. 5: 238–243. https://doi.org/10.1002/pfi.4170300510.

Silk, E. M., A. E. Rechkemmer, S. R. Daly, K. W. Jablokow, and S. McKilligan. 2021. "Problem Framing and Cognitive Style: Impacts on Design Ideation Perceptions." *Design Studies* 74: 101015. https://doi.org/10.1016/j.destud.2021.101015.

Slater, S. F., J. J. Mohr, and S. Sengupta. 2014. "Radical Product Innovation Capability: Literature Review, Synthesis, and Illustrative Research Propositions." *Journal of Product Innovation Management* 31, no. 3: 552–566. https://doi.org/10.1111/jpim.12113.

Snow, D. A., and R. D. Benford. 1988. "Ideology, Frame Resonance, and Participant Mobilization." *International Social Movement Research* 1, no. 1: 197–217.

Stierand, M., and V. Dörfler. 2016. "The Role of Intuition in the Creative Process of Expert Chefs." *Journal of Creative Behavior* 50, no. 3: 178–185. https://doi.org/10.1002/jocb.100.

Stigliani, I., and D. Ravasi. 2012. "Organizing Thoughts and Connecting Brains: Material Practices and the Transition From Individual to Group-Level Prospective Sensemaking." *Academy of Management Journal* 55, no. 5: 1232–1259. https://doi.org/10.5465/amj.2010.0890.

Stompff, G., F. Smulders, and L. Henze. 2016. "Surprises Are the Benefits: Reframing in Multidisciplinary Design Teams." *Design Studies* 47: 187–214. https://doi.org/10.1016/j.destud.2016.09.004.

Stumpf, S. C., and J. T. McDonnell. 2002. "Talking About Team Framing: Using Argumentation to Analyse and Support Experiential Learning in Early Design Episodes." *Design Studies* 23, no. 1: 5–23. https://doi.org/10.1016/S0142-694X(01)00020-5.

Sukhov, A., A. Sihvonen, J. Netz, P. Magnusson, and L. E. Olsson. 2021. "How Experts Screen Ideas: The Complex Interplay of Intuition, Analysis and Sensemaking." *Journal of Product Innovation Management* 38, no. 2: 248–270. https://doi.org/10.1111/jpim.12559.

Sukhov, A., A. Sihvonen, L. E. Olsson, and P. R. Magnusson. 2018. "That Makes Sense to Me: Openness to Change and Sensemaking in Idea Screening." *International Journal of Innovation Management* 22, no. 8: 1840009. https://doi.org/10.1142/S1363919618400091.

Tsai, F.-S. 2023. "When and How Group Diversity Facilitate Innovativeness? The Roles of Knowledge Heterogeneity and Governance." *Knowledge Management Research and Practice* 21, no. 3: 566–576. https://doi.org/10.1080/14778238.2021.2004950.

Tsoukas, H. 2009. "A Dialogical Approach to the Creation of New Knowledge in Organizations." *Organization Science* 20, no. 6: 941–957.

Ulrich, F., S. Mengiste, and S. Müller. 2015. "Informal Evaluation and Institutionalization of Neoteric Technology Ideas: The Case of Two Danish Organizations." *Communications of the Association for Information Systems* 37: 949–974. https://doi.org/10.17705/1CAIS.03747.

Unsworth, K. 2001. "Unpacking Creativity." *Academy of Management Review* 26, no. 2: 289–297. https://doi.org/10.5465/amr.2001.4378025.

Vaccaro, A., S. Brusoni, and F. M. Veloso. 2011. "Virtual Design, Problem Framing, and Innovation: An Empirical Study in the Automotive

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Industry." *Journal of Management Studies* 48, no. 1: 99–122. https://doi.org/10.1111/j.1467-6486.2010.00939.x.

Van Leeuwen, J. P., D. Rijken, I. Bloothoofd, and E. Cobussen. 2020. "Finding New Perspectives Through Theme Investigation." *Design Journal* 23, no. 3: 441–461. https://doi.org/10.1080/14606925.2020.1744258.

Van Maanen, J. 1979. "The Fact of Fiction in Organizational Ethnography." *Administrative Science Quarterly* 24, no. 4: 539–550. https://doi.org/10.2307/2392360.

van Maanen, J. 2011. *Tales of the Field: On Writing Ethnography*. second ed. University of Chicago Press.

Verganti, R. 2009. *Design Driven Innovation: Changing the Rules of Competition by Radically Innovating What Things Mean*. Harvard Business Press.

Verganti, R. 2016. "The Innovative Power of Criticism." *Harvard Business Review* 94, no. 1: 18.

Verganti, R. 2017. *Overcrowded: Designing Meaningful Products in a World Awash With Ideas*. MIT Press.

Verganti, R., C. Dell'Era, and K. S. Swan. 2021. "Design Thinking: Critical Analysis and Future Evolution." *Journal of Product Innovation Management* 38, no. 6: 603–622. https://doi.org/10.1111/jpim.12610.

Verganti, R., and Å. Öberg. 2013. "Interpreting and Envisioning — A Hermeneutic Framework to Look at Radical Innovation of Meanings." *Industrial Marketing Management* 42, no. 1: 86–95. https://doi.org/10.1016/j.indmarman.2012.11.012.

Verganti, R., and A. B. R. Shani. 2016. "Vision Transformation Through Radical Circles." *Organizational Dynamics* 2: 104–113. https://doi.org/10.1016/j.orgdyn.2016.02.004.

Wacquant, L. 2002. "The Curious Eclipse of Prison Ethnography in the Age of Mass Incarceration." *Ethnography* 3, no. 4: 371–397. https://doi.org/10.1177/1466138102003004012.

Weick, K. 1995. *Sensemaking in Organizations*. Vol. 3. Sage Publications.

Weick, K. E., K. M. Sutcliffe, and D. Obstfeld. 2005. "Organizing and the Process of Sensemaking." *Organization Science* 16, no. 4: 409–421. https://doi.org/10.1287/orsc.1050.0133.

Wyrtki, K., M. Röglinger, and M. Rosemann. 2021. "Opportunity-Led Ideation: How to Convert Corporate Opportunities Into Innovative Ideas." *Creativity and Innovation Management* 30, no. 3: 523–541. https://doi.org/10.1111/caim.12450.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supplementary Information.

15405885, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jpim.70038, Wiley Online Library on [13/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License